Aspirin (Lexi-Drugs Multinational)

Monograph Images Adult Patient Education Pediatric Patient Education

Brand Names: International

Find brand name(s) by country (for United States and Canada see separate Brand Names sections)

Enter a Country or Country Code

For country code abbreviations (show table)

Brazilian Nonproprietary Names (DCB)

Ácido Acetilsalicílico

Japanese Accepted Name (JAN)

アスピリン

Anatomic Therapeutic Chemical (ATC) Classification

- B01AC06
- N02BA01
- N02BA51

Pharmacologic Category

Analgesic, Nonopioid; Antiplatelet Agent; Nonsteroidal Anti-inflammatory Drug (NSAID), Oral; Salicylate

Dosing: Adult

Dosage guidance:

Safety: Avoid regular or frequent use of nonsteroidal anti-inflammatory drugs (NSAIDs) in patients taking aspirin for cardiovascular protection as these may reduce the cardioprotective effects of aspirin (Capone 2005; Catella-Lawson 2001; MacDonald 2003).

Dosing: Dosing information below is based on the IR formulations; do not use the ER formulation in situations when a rapid onset of action is necessary.

Clinical considerations: In patients who require concomitant therapeutic anticoagulation, antiplatelet selection and/or duration of therapy may differ in order to balance risks for thrombosis and bleeding (ACC [Kumbhani 2021]).

Anti-inflammatory for arthritis associated with rheumatic disease

Anti-inflammatory for arthritis associated with rheumatic disease: Immediate release: Oral: 4 to 8 g/day in 4 to 5 divided doses as needed; titrate dose based on response and tolerability. Continue treatment until symptoms resolve (typically 1 to 2 weeks, but potentially up to 8 weeks). Use of aspirin at these high doses (4 to 8 g/day) may be limited by adverse effects (tinnitus, diminished auditory acuity, GI intolerance), making other available NSAIDs preferred (Abramson 2021; Carapetis 2012; Steer 2019).

Atherosclerotic cardiovascular disease

Atherosclerotic cardiovascular disease:

Acute coronary syndrome:

Note: For rapid onset, non–enteric-coated IR tablet(s) should be chewed and swallowed upon identification of clinical and ECG findings suggesting an acute coronary syndrome. Enteric-coated aspirin is not preferred, since onset of action may be delayed. If it is the only product available, enteric-coated IR tablet(s) may be chewed and swallowed (ACCP [Eikelboom 2012]; Sai 2011). For maintenance therapy, any oral formulation is acceptable for use.

Non–ST-elevation acute coronary syndromes or ST-elevation myocardial infarction:

Note: For initial therapy, administer aspirin in combination with an IV anticoagulant and a $P2Y_{12}$ inhibitor (ACC/AHA [Lawton 2022]).

Initial:

Immediate release (non–enteric-coated): Oral: 162 to 325 mg administered once (chew and swallow) at the time of diagnosis (ACC/AHA [Lawton 2022]).

Rectal (alternative route): 600 mg administered once at the time of diagnosis if an IR oral formulation is unavailable or oral route is not feasible (Maalouf 2009).

Maintenance (secondary prevention):

Immediate release: Oral: 75 to 100 mg once daily (ACC/AHA [Lawton 2022]; Mehta 2001).

Duration of therapy:

Preferred approach: Aspirin plus a $P2Y_{12}$ inhibitor (dual antiplatelet therapy [DAPT]) should be continued for ≥12 months, unless major bleeding is a concern. For patients at high risk of bleeding or who experience overt bleeding, DAPT for 6 months may be reasonable. If there have been no major bleeding complications after 12 months, continuation of DAPT may be considered. Reevaluate the need for DAPT at regular intervals based on bleeding and thrombotic risks. When DAPT is complete, discontinue the $P2Y_{12}$ inhibitor and continue aspirin indefinitely for secondary prevention (ACC/AHA [Lawton 2022]; Bonaca 2015; Mauri 2014; Mehta 2001; Wallentin 2009; Wiviott 2007; Yusuf 2001).

Alternative approach in select patients to minimize bleeding events: Aspirin in combination with a $P2Y_{12}$ inhibitor (DAPT) should be continued for 1 to 3 months after PCI, then discontinue aspirin and continue $P2Y_{12}$ inhibitor monotherapy. When the $P2Y_{12}$ inhibitor is discontinued, restart aspirin for secondary prevention (ACC/AHA [Lawton 2022]; Hahn 2019; Kim 2020; Mehran 2019; O'Donoghue 2020; Vranckx 2018; Watanabe 2019).

Percutaneous coronary intervention for stable ischemic heart disease (off-label use):

Initial:

Note: For initial therapy, non–enteric-coated IR tablet(s) should be administered. Enteric-coated aspirin is not preferred since onset of action is delayed. For patients who receive a coronary stent during percutaneous coronary intervention, administer aspirin in combination with a parenteral anticoagulant and a $P2Y_{12}$ inhibitor (clopidogrel or ticagrelor; prasugrel is **not** recommended in this patient population) (ACC/AHA [Lawton 2022]).

Immediate release (non–enteric-coated): Oral: 162 to 325 mg given ≥2 hours (preferably 24 hours) before the procedure (ACC/AHA [Lawton 2022]).

Maintenance:

Immediate release: Oral: 75 to 100 mg once daily in combination with clopidogrel **or** ticagrelor (DAPT) (ACC/AHA [Lawton 2022]). Refer to clopidogrel or ticagrelor monograph for information on duration of DAPT and subsequent management.

Atherosclerotic cardiovascular disease, primary prevention (off-label use):

Note: Use should be a shared decision between health care professionals and patients ≥40 years of age who are at increased risk for cardiovascular disease, after weighing risks versus benefits. Some experts recommend against this use for patients ≥60 years of age (USPSTF 2022), while others recommend against this use for patients >70 years of age (ACC/AHA [Arnett 2019]).

Immediate release: Oral: 75 to 100 mg once daily (ACC/AHA [Arnett 2019]).

Atherosclerotic cardiovascular disease, secondary prevention:

Carotid artery atherosclerosis, asymptomatic or symptomatic (off-label use): Immediate release: Oral: 75 to 325 mg once daily (ACCP [Alonso-Coello 2012]; Walker 1995).

Coronary artery bypass graft surgery: Immediate release: Oral: 75 to 81 mg once daily beginning preoperatively; continue indefinitely following surgery (AHA [Kulik 2015]; Aranki 2021). Following surgery, some experts restart aspirin (with or without a one-time, 325 mg loading dose) approximately 6 hours after completion of the procedure or after extubation, whichever comes first (Aranki 2021).

Off-pump coronary artery bypass graft surgery:Following surgery, consider adding clopidogrel in combination with aspirin for 12 months then discontinue clopidogrel and continue aspirin indefinitely (AHA [Kulik 2015]).

Patients with acute coronary syndrome followed by coronary artery bypass graft surgery: Administer aspirin in combination with a $P2Y_{12}$ inhibitor for 12 months then continue aspirin indefinitely (AHA [Kulik 2015]). Some experts do not use $P2Y_{12}$ inhibitors postoperatively in these patients (Aranki 2021).

Ischemic stroke/transient ischemic attack:

Note: In patients who receive IV thrombolytic therapy, antiplatelet therapy is generally delayed for at least 24 hours, but administered as soon as possible thereafter (AHA/ASA [Powers 2019]; Oliveira-Filho 2022).

Due to intracranial large artery atherosclerosis (50% to 99%), secondary prevention:

Note: Avoid dual antiplatelet therapy (DAPT) in patients with hemorrhagic transformation (Oliveira-Filho 2022).

Immediate release: Oral:

Monotherapy: 325 mg once daily. May consider a short course of DAPT as outlined below. (AHA/ASA [Kleindorfer 2021]).

Short-term (90 days) DAPT with concomitant clopidogrel in patients with recent (≤30 days) stroke or TIA due to 70% to 99% stenosis of an intracranial large artery: 325 mg once daily followed by long-term aspirin monotherapy (AHA/ASA [Kleindorfer 2021]; Turan 2024).

Short-term (21 days) DAPT with concomitant clopidogrel in patients with recent (≤30 days) minor stroke or high-risk TIA due to 50% to 69% stenosis of an intracranial large artery: 160 to 325 mg loading dose followed by 50 to 100 mg once daily (Turan 2024).

Short-term (21 to 30 days) DAPT with concomitant ticagrelor in patients with recent (≤24 hours) minor stroke or high-risk TIA and >30% stenosis of ipsilateral major intracranial artery: 300 to 325 mg loading dose followed by 75 to 100 mg once daily (AHA/ASA [Kleindorfer 2021]; Oliveira-Filho 2022; ticagrelor manufacturer's labeling).

Due to other noncardioembolic causes (eg, small vessel disease), secondary prevention:

Note: For patients with recent (eg, ≤24 hours) minor stroke or high-risk TIA, may consider short-term DAPT with concomitant clopidogrel (21 to 90 days) or ticagrelor (21 to 30 days) followed by long-term single-agent antiplatelet therapy with aspirin, clopidogrel, or aspirin/ER dipyridamole. Avoid DAPT in patients with hemorrhagic transformation (AHA/ASA [Kleindorfer 2021]; Oliveira-Filho 2022).

Initial:

Immediate release: Oral: 160 to 325 mg administered once at the time of diagnosis (AHA/ASA [Powers 2019]; Oliveira-Filho 2022).

Rectal (alternative route): 300 mg administered once at the time of diagnosis if oral route is not feasible (AHA/ASA [Powers 2019]; IST 1997; Sandercock 2014).

Maintenance:

Immediate release: Oral: 50 to 325 mg once daily; some experts recommend 50 to 100 mg once daily when used with clopidogrel for DAPT and as long-term single-agent therapy (ACCP [Lansberg 2012]; AHA/ASA [Kleindorfer 2021]; Cucchiara 2022). When used with ticagrelor for DAPT, 75 to 100 mg once daily is recommended (ticagrelor manufacturer's labeling).

Peripheral atherosclerotic disease (upper or lower extremity; with or without a revascularization procedure) (off-label use): Immediate release: Oral: 75 to 100 mg once daily (ACCP [Alonso-Coello 2012]; AHA/ACC [Gerhard-Herman 2017]).

Stable ischemic heart disease: Immediate release: Oral: 75 to 100 mg once daily (AHA/ACC [Virani 2023]; Kannam 2019).

Carotid artery stenting

Carotid artery stenting (off-label use):

Percutaneous approach:

Initial:

***Initiation ≥48 hours before procedure*: Immediate release: Oral:** 325 mg twice daily in combination with clopidogrel (or ticagrelor if clopidogrel cannot be used) (Brott 2010, Jim 2024a, Marcaccio 2022).

***Initiation <48 hours before procedure*: Immediate release: Oral:** 650 mg once at least 4 hours before procedure in combination with clopidogrel (or ticagrelor if clopidogrel cannot be used) (Brott 2010, Jim 2024a, Marcaccio 2022).

Maintenance:

Immediate release: Oral: 75 to 325 mg once daily in combination with clopidogrel (or ticagrelor if clopidogrel cannot be used) for at least 4 weeks, then discontinue clopidogrel and continue aspirin 75 to 325 mg once daily indefinitely thereafter. In patients with history of neck irradiation, some experts recommend continuing aspirin plus clopidogrel indefinitely (ASA/ACCF/AHA [Brott 2011]; Brott 2010; Jim 2024a, Marcaccio 2022).

Transcarotid approach:

Initial:

***Initiation ≥72 hours before procedure:* Immediate release: Oral:** 75 to 325 mg once daily in combination with clopidogrel (or ticagrelor if clopidogrel cannot be used) (Jim 2024b, Kwolek 2015, Marcaccio 2022).

***Initiation <72 hours before procedure*: Immediate release: Oral:** 650 mg once at least 4 hours before procedure in combination with clopidogrel (or ticagrelor if clopidogrel cannot be used) (Jim 2024b, Kwolek 2015, Marcaccio 2022).

Maintenance:

Immediate release: Oral: 75 to 325 mg once daily in combination with clopidogrel (or ticagrelor if clopidogrel cannot be used) for at least 4 weeks, then discontinue clopidogrel and continue aspirin 75 to 325 mg once daily indefinitely thereafter. In patients with history of neck irradiation, some experts recommend continuing aspirin plus clopidogrel indefinitely (Brott 2010; Jim 2024b; Kwolek 2015, Marcaccio 2022).

Carotid endarterectomy

Carotid endarterectomy: Immediate release: Oral: 75 to 325 mg once daily starting prior to surgery and continued indefinitely (ACCP [Alonso-Coello 2012]; Fairman 2021).

Colorectal cancer risk reduction, primary prevention

Colorectal cancer risk reduction, primary prevention (off-label use):

Note: The optimal dose and duration of therapy for colorectal cancer risk reduction are unknown. Utilization should be a shared decision between health care professionals and patients that weighs the risk versus benefits of treatment (Chan 2020).

Immediate release: Oral: 75 to 325 mg once daily (Chan 2020; Rothwell 2010; Ye 2013).

Migraine, acute treatment

Migraine, acute treatment (off-label use):

Note: For mild to moderate attacks not associated with vomiting or severe nausea (Schwedt 2021).

Immediate release: Oral: 900 mg or 1 g once (Lipton 2005; MacGregor 2002).

Pain or fever

Pain or fever:

Immediate release:

Oral: 325 mg to 1 g every 4 to 6 hours as needed; usual maximum daily dose: 4 g/day (Abramson 2021).

Note: If patient cannot take orally, rectal suppositories (300 or 600 mg) are available.

IM/IV [International product]: Note: Dose is expressed as acetylsalicylic acid (900 mg lysine acetylsalicylate is equivalent to 500 mg acetylsalicylic acid [aspirin]): 500 mg to 1 g every 4 hours as needed; maximum: 3 g/day.

Note: Reserve IM/IV administration for severe pain in patients who are unable to tolerate oral formulations.

Pericarditis, acute or recurrent

Pericarditis, acute or recurrent (off-label use):

Note: Aspirin is preferred over other nonsteroidal anti-inflammatory drugs (NSAIDs) in patients with ischemic heart disease. If pericarditis occurs secondary to acute myocardial infarction (MI), avoid NSAIDs (other than once-daily, low-dose aspirin for treatment of MI) and use acetaminophen for analgesia. If this does not suffice or if additional pericarditis treatment is necessary after 7 to 10 days, aspirin is preferred and other NSAIDs should continue to be avoided (ACCF/AHA [O'Gara 2013]; LeWinter 2022).

Immediate release: Oral: Initial: 650 mg to 1 g every 8 hours until resolution of symptoms for at least 24 hours and normalization of inflammatory biomarkers (eg, C-reactive protein) if monitored; initial therapy typically lasts for ≥1 to 2 weeks. Gradually taper over several weeks by decreasing each dose by 250 to 500 mg every 1 to 2 weeks; during taper, ensure patient remains asymptomatic and inflammatory biomarkers remain normal (if monitored). Use in combination with colchicine. In patients at risk of NSAID-related GI toxicity, prophylaxis (generally with a proton pump inhibitor) is recommended (ESC [Adler 2015]; Imazio 2022).

Polycythemia vera, prevention of thrombosis

Polycythemia vera, prevention of thrombosis (off-label use):

Note: Use with caution in patients with concurrent acquired von Willebrand syndrome due to risk for bleeding (Tefferi 2024).

Immediate release: Oral: 75 to 100 mg once daily (Barbui 2006; BSH [McMullin 2019]; Landolfi 2004; Tefferi 2024).

Preeclampsia prevention

Preeclampsia prevention (off-label use):

Note: Consider for use in pregnant women with ≥2 moderate risk factors or ≥1 high risk factor for preeclampsia (ACOG 743 2018).

Immediate release: Oral: 81 to 162 mg once daily, ideally beginning between 12 to 16 weeks' gestation but may be started up to 28 weeks' gestation; continue therapy until delivery (ACOG 743 2018; Rolnik 2017).

Valvular heart disease

Valvular heart disease:

Surgical prosthetic heart valve replacement, thromboprophylaxis:

Bioprosthetic aortic or mitral heart valve replacement (off-label use): Note: Timing of aspirin initiation is based on bleeding risk. For patients at low risk of bleeding, treat with warfarin monotherapy for 3 to 6 months, then transition to aspirin monotherapy. For patients at elevated risk of bleeding, initiate aspirin within 24 hours of surgery (ie, no initial treatment with warfarin) (ACC/AHA [Otto 2021]; Nkomo 2022).

Immediate release: Oral: 75 to 100 mg once daily (ACC/AHA [Otto 2021]).

Mechanical aortic or mitral heart valve replacement (off-label use): Note: Aspirin is required, in combination with warfarin, for patients who receive an On-X mechanical aortic valve. For other mechanical valve types, consider risks of thromboembolism and bleeding before adding aspirin; warfarin monotherapy is typically sufficient. Addition of aspirin is not routinely required in the absence of another indication for antiplatelet therapy (ACC/AHA [Otto 2021]; Puskas 2014).

Immediate release: Oral: 75 to 100 mg once daily in combination with warfarin (ACC/AHA [Otto 2021]; Puskas 2014).

Transcatheter *aortic* valve replacement, thromboprophylaxis (off-label use):

Note: Refer to institutional policies and procedures on use of antiplatelet therapy for patients who require therapeutic anticoagulation for a different indication.

Immediate release: Oral: 75 to 100 mg once daily; may use in combination with clopidogrel for 3 to 6 months after transcatheter aortic valve replacement, depending on type of valve implanted (ACC/AHA [Otto 2021]). To minimize risk of bleeding complications, may give aspirin **or** clopidogrel alone and reserve dual antiplatelet therapy during the first 3 to 6 months for patients at high risk of a thrombotic event; for either strategy, continue aspirin indefinitely after the initial 3 to 6 months of therapy (Brouwer 2020; Kuno 2019).

Transcatheter *mitral* valve repair with MitraClip device, thromboprophylaxis (off-label use):

Note: Patients are generally treated with antithrombotic therapy (antiplatelet or anticoagulant if there is a concurrent indication) for at least 6 months following the procedure.

Immediate release: Oral:

Loading dose: 325 mg once immediately following MitraClip insertion or within 24 hours prior to the procedure; use in combination with clopidogrel (Stone 2018).

Maintenance: 81 mg once daily for at least 6 months; may use as monotherapy or in combination with clopidogrel (Stone 2018).

Venous thromboembolism prevention, indefinite therapy

Venous thromboembolism prevention, indefinite therapy (off-label use):

Note: For use in select patients to prevent recurrent venous thromboembolism (VTE) if unable to take an anticoagulant. In patients who have completed ≥6 months of anticoagulation and in whom indefinite therapeutic anticoagulation is indicated, aspirin is not recommended since it is less effective (ACCP [Stevens 2021]).

Immediate release: Oral: 100 mg once daily after completion of a conventional treatment course with therapeutic anticoagulation (Becattini 2012; Brighton 2012; Simes 2014).

Venous thromboembolism prophylaxis for total hip or total knee arthroplasty

Venous thromboembolism prophylaxis for total hip or total knee arthroplasty (off-label use):

Note: Some experts limit this strategy to lower-risk patients who undergo elective unilateral total hip arthroplasty (THA) or total knee arthroplasty (TKA), ambulate within 24 hours after surgery, and do not have additional risk factors for VTE, indications for long-term anticoagulation, lower limb or hip fracture in the previous 3 months, or expected major surgery in the upcoming 3 months (Douketis 2023).

Immediate release: Oral: Begin postoperative prophylaxis with rivaroxaban or a low-molecular-weight heparin, then discontinue therapy after 5 days for TKA or after 5 to 10 days for THA, and initiate aspirin 81 mg once daily. For TKA, continue aspirin through postoperative day 10 to 14 (if not fully ambulatory by day 14, may continue through postoperative day 35). For THA, continue aspirin through postoperative day 35 (Anderson 2013; Anderson 2018; Douketis 2023).

Venous thromboembolism prophylaxis in lower-risk patients with multiple myeloma receiving immunomodulatory therapy

Venous thromboembolism prophylaxis in lower-risk patients with multiple myeloma receiving immunomodulatory therapy (eg, lenalidomide, pomalidomide, thalidomide) (off-label use): Oral: 81 to 325 mg once daily during immunomodulatory therapy (Larocca 2012; Palumbo 2011; Rajkumar 2005).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

* See Dosage and Administration in AHFS Essentials for additional information.

Dosing: Older Adult

IM/IV [International product]: Analgesic and antipyretic: Note: Dose is expressed as acetylsalicylic acid (900 mg lysine acetylsalicylate is equivalent to 500 mg acetylsalicylic acid [aspirin]): 500 mg to 1 g every 4 hours as needed; maximum: 2 g/day.

Note: Reserve IM/IV administration for severe pain in patients who are unable to tolerate oral formulations.

Oral, rectal: Refer to adult dosing. Avoid for the primary prevention of cardiovascular disease. In older adults already taking for primary prevention, consider deprescribing. Unless alternative agents are ineffective and a gastroprotective agent can be administered, avoid short-term scheduled use of doses >325 mg in combination with corticosteroids, anticoagulants, or antiplatelet agents or chronic use with or without medications that increase risk for bleeding (Beers Criteria [AGS 2023]).

Dosing: Altered Kidney Function: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS

Antiplatelet uses:

Note: In general the benefit of low-dose aspirin outweighs any risk associated with nephropathy or other adverse effects even in the setting of severe kidney impairment; the recommended aspirin dose should not be reduced in any patient with suspected or documented cardiovascular disease, or other antithrombotic indication ([Ref]).

Altered kidney function: No dosage adjustment necessary for any degree of kidney impairment ([Ref]).

Hemodialysis, intermittent (thrice weekly): No dosage adjustment necessary ([Ref]).

Peritoneal dialysis: No dosage adjustment necessary ([Ref]).

CRRT: No dosage adjustment necessary ([Ref]).

PIRRT (eg, sustained, low-efficiency diafiltration): No dosage adjustment necessary ([Ref]).

Analgesia or anti-inflammatory uses:

Altered kidney function:

CrCl >10 mL/minute: No dosage adjustment necessary; however, high doses have been associated with acute kidney injury (AKI) ([Ref]). Use lowest effective dosage and limit duration of therapy, particularly for patients at high risk for developing AKI (eg, patients with chronic kidney disease, volume depletion, older age) ([Ref]).

CrCl <10 mL/minute: Avoid use (manufacturer's labeling; expert opinion). May exacerbate uremic GI and hematologic symptoms ([Ref]).

Hemodialysis, intermittent (thrice weekly): Avoid use. May exacerbate uremic GI and hematologic symptoms ([Ref]).

Peritoneal dialysis: Avoid use. May exacerbate uremic GI and hematologic symptoms ([Ref]).

CRRT: Avoid use ([Ref]).

PIRRT (eg, sustained, low-efficiency diafiltration): Avoid use ([Ref]).

Dosing: Altered Liver Function: Adult

Avoid use in severe liver disease.

Dosing: Pediatric

Dosage guidance:

Dosing: Two immediate-release formulations are available: tablets and liquid-filled capsules (Vazalore). Immediate-release capsules cannot be divided and may not be an appropriate dosage form for some pediatric patients. All pediatric dosing for immediate-release formulations unless otherwise specified. Doses are typically rounded to a convenient amount (eg, $^{1}/_{4}$ of 81 mg tablet).

Analgesic

Analgesic: Oral, rectal: **Note:** Do not use aspirin in pediatric patients <18 years who have or who are recovering from chickenpox or flu symptoms (eg, viral illness) due to the association with Reye syndrome ([Ref]):

Infants, Children, and Adolescents weighing <50 kg: Limited data available: 10 to 15 mg/kg/dose every 4 to 6 hours; maximum daily dose: 90 mg/kg/**day** or 4,000 mg/**day** whichever is less ([Ref]).

Children ≥12 years and Adolescents weighing ≥50 kg: 325 to 650 mg every 4 to 6 hours; maximum daily dose: 4,000 mg/**day** ([Ref]).

Anti-inflammatory

Anti-inflammatory: Limited data available: Infants, Children, and Adolescents: Oral: Initial: 60 to 90 mg/kg/**day** in divided doses; usual maintenance: 80 to 100 mg/kg/**day** divided every 6 to 8 hours; monitor serum concentrations ([Ref]). **Note:** Although included in some product labeling, the use of salicylates including aspirin for the treatment of juvenile idiopathic arthritis has been replaced by other pharmacologic agents with improved efficacy and adverse effect profiles (eg, nonsalicylate nonsteroidal anti-inflammatory drugs, disease-modifying antirheumatic drugs, glucocorticoids, biologic modifiers) ([Ref]).

Antiplatelet effect

Antiplatelet effect: Limited data available: Infants, Children, and Adolescents: Oral: Adequate pediatric studies have not been performed; pediatric dosage is derived from adult studies. Usual adult maximum daily dose for antiplatelet effects is 325 mg/**day**.

Acute ischemic stroke (AIS):

Noncardioembolic: 1 to 5 mg/kg/dose once daily for ≥2 years; patients with recurrent AIS or TIAs should be transitioned to clopidogrel, LMWH, or warfarin ([Ref]).

Secondary to Moyamoya and non-Moyamoya vasculopathy: 1 to 5 mg/kg/dose once daily; **Note:** In non-Moyamoya vasculopathy, continue aspirin for 3 months, with subsequent use guided by repeat cerebrovascular imaging ([Ref]).

Prosthetic heart valve:

Bioprosthetic aortic valve (with normal sinus rhythm): 1 to 5 mg/kg/dose once daily for 3 months ([Ref]).

Mechanical aortic and/or mitral valve: 1 to 5 mg/kg/dose once daily combined with vitamin K antagonist (eg, warfarin) is recommended as first-line antithrombotic therapy ([Ref]). Alternative regimens: 6 to 20 mg/kg/dose once daily in combination with dipyridamole ([Ref]).

Shunts: Blalock-Taussig; Glenn; postoperative; primary prophylaxis: 1 to 5 mg/kg/dose once daily ([Ref]).

Norwood, Fontan surgery, postoperative; primary prophylaxis: 1 to 5 mg/kg/dose once daily ([Ref]).

Transcatheter Atrial Septal Defect (ASD) or Ventricular Septal Defect (VSD) devices, postprocedure prophylaxis: 1 to 5 mg/kg/dose once daily starting one to several days prior to implantation and continued for at least 6 months. For older children and adolescents, after device closure of ASD, an additional anticoagulant may be given with aspirin for 3 to 6 months, but the aspirin should continue for at least 6 months ([Ref]).

Ventricular assist device (VAD) placement: 1 to 5 mg/kg/dose once daily initiated within 72 hours of VAD placement; should be used with heparin (initiated between 8 to 48 hours following implantation) and with or without dipyridamole ([Ref]).

Kawasaki disease, treatment

Kawasaki disease, treatment: Limited data available: **Note:** Patients with Kawasaki disease and presenting with influenza or viral illness should not receive aspirin; acetaminophen is suggested as an alternate antipyretic in these patients and an alternate antiplatelet agent is suggested for a minimum of 2 weeks ([Ref]).

Initial therapy (acute phase): **Note:** Optimal dose not established. Use in combination with IV immune globulin.

Infants and Children:

Moderate dose: Oral: 30 to 50 mg/kg/**day** divided every 6 hours until fever resolves for at least 48 to 72 hours ([Ref]).

High dose: Oral: 80 to 100 mg/kg/**day** divided every 6 hours until fever resolves for at least 48 to 72 hours ([Ref]).

Subsequent therapy (low-dose; antiplatelet effects): Infants and Children: Oral: 3 to 5 mg/kg/**day** once daily; initiate at least 48 to 72 hours after resolution of fever and continue for 6 to 8 weeks in patients without coronary artery abnormalities. In patients with coronary artery abnormalities ranging from dilation to large/giant aneurysms, low-dose aspirin should be continued (with or without dual antiplatelet therapy, warfarin or low molecular weight heparin); duration of aspirin therapy is based upon severity of coronary artery involvement; recommend consulting published treatment and institutional guidelines, and expert medical consultation for additional guidance ([Ref]).

Multisystem inflammatory syndrome in children associated with SARS-CoV-2

Multisystem inflammatory syndrome in children (MIS-C) associated with SARS-CoV-2 (antiplatelet/antithrombotic effects): Limited data available:

Note: Recommended for all MIS-C patients without active bleeding, significant bleeding risk, or platelet count ≤80,000 cells/mm^3 ([Ref]). Use is based on limited experience in patients with MIS-C, emerging data in adults with COVID-19, and extrapolation from Kawasaki disease and myocarditis ([Ref]).

Infants, Children, and Adolescents: Oral: 3 to 5 mg/kg/**day** once daily; maximum daily dose: 81 mg/**day**; continue until normalization of platelet count, confirmed normal coronary arteries at ≥4 weeks after diagnosis, and ejection fraction >35% ([Ref]).

Rheumatic fever

Rheumatic fever: Limited data available: Infants, Children, and Adolescents: Oral: Initial: 100 mg/kg/**day** divided into 4 to 5 doses; if response inadequate, may increase dose to 125 mg/kg/**day**; continue for 2 weeks; then decrease dose to 60 to 70 mg/kg/**day** in divided doses for an additional 3 to 6 weeks ([Ref]).

Migratory polyarthritis, with carditis without cardiomegaly or congestive heart failure: Oral: Initial: 50 to 70 mg/kg/**day** in 4 divided doses for 3 to 5 days, followed by 50 mg/kg/**day** in 4 divided doses for 2 to 3 weeks, followed by 25 mg/kg/**day** in 4 divided doses for 2 to 4 weeks ([Ref]); escalation to doses up to 80 to 100 mg/kg/**day** in 4 or 5 divided doses has been described for arthritis management ([Ref]).

Carditis and more than minimal cardiomegaly or congestive heart failure: **Note:** Aspirin should be initiated at the beginning of prednisone taper regimen to prevent rebound inflammation: Oral: 50 mg/kg/**day** in 4 divided doses for 6 weeks ([Ref]).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Altered Kidney Function: Pediatric

Infants, Children, and Adolescents: There are no recommendations in the manufacturer's labeling; however, the following adjustments have been recommended ([Ref]):

GFR ≥10 mL/minute/1.73 m^2: No dosage adjustment necessary.

GFR <10 mL/minute/1.73 m^2: Avoid use.

Intermittent hemodialysis: Dialyzable: 50% to 100% (concentration dependent; higher salicylate concentrations are more readily dialyzable) ([Ref]); administer daily dose after dialysis session on dialysis days ([Ref]).

Peritoneal dialysis: Avoid use.

CRRT: No dosage adjustment necessary; monitor serum concentrations.

Dosing: Altered Liver Function: Pediatric

All ages: Avoid use in severe liver disease.

Calculations

- Creatinine Clearance by Cockcroft-Gault
- Creatinine Clearance by Cockcroft-Gault (SI units)
- Creatinine Clearance by Cockcroft-Gault with IBW
- Creatinine Clearance by Cockcroft-Gault with IBW (SI units)

Use: Labeled Indications

Oral:

Immediate release:

Analgesic, antipyretic, and anti-inflammatory: For the temporary relief of headache, pain, and fever caused by colds, muscle aches and pains, menstrual pain, toothache pain, and minor aches and pains of arthritis.

Revascularization procedures: For use in patients who have undergone revascularization procedures (ie, coronary artery bypass graft, percutaneous transluminal coronary angioplasty, or carotid endarterectomy).

Vascular indications, including ischemic stroke, transient ischemic attack, acute coronary syndromes (ST-elevation myocardial infarction or non-ST-elevation acute coronary syndromes [non-ST-elevation myocardial infarction or unstable angina]), secondary prevention after acute coronary syndromes, and management of stable ischemic heart disease: To reduce the combined risk of death and nonfatal stroke in patients who have had ischemic stroke or transient ischemia of the brain due to fibrin platelet emboli; to reduce the risk of vascular mortality in patients with a suspected acute myocardial infarction (MI); to reduce the combined risk of death and nonfatal MI in patients with a previous MI or unstable angina; to reduce the combined risk of MI and sudden death in patients with stable ischemic heart disease.

ER capsules:

Ischemic stroke or transient ischemic attack: To reduce the risk of death and recurrent stroke in patients who have had an ischemic stroke or transient ischemic attack.

Stable ischemic heart disease: To reduce the risk of death and MI in patients with stable ischemic heart disease.

Limitations of use: Do not use ER capsules in situations for which a rapid onset of action is required (such as acute treatment of MI or before percutaneous coronary intervention); use IR formulations instead.

Injection [International product]: Analgesic and antipyretic: Treatment of severe pain and relief of fever symptoms in adult patients unable to take oral NSAIDs.

* See Uses in AHFS Essentials for additional information.

Use: Off-Label: Adult

Atherosclerotic cardiovascular disease, primary prevention Level of Evidence [G]

Based on the American College of Cardiology/American Heart Association (ACC/AHA) guideline on the primary prevention of cardiovascular disease and the American Diabetes Association standards of care in diabetes, aspirin may be used for the primary prevention of cardiovascular disease in select patients after weighing the cardiovascular disease risk versus benefits ([Ref]).

Carotid artery atherosclerosis, asymptomatic or symptomatic Level of Evidence [G]

Based on the American College of Chest Physicians (ACCP) guidelines for antithrombotic therapy and prevention of thrombosis (9th edition), daily aspirin is suggested in patients with asymptomatic or symptomatic carotid artery atherosclerosis based on a slight reduction in total mortality observed when aspirin is taken over 10 years (regardless of cardiovascular risk profile) ([Ref]). The AHA/American Stroke Association guidelines for the primary prevention of stroke recommend daily aspirin for patients with asymptomatic or symptomatic carotid atherosclerosis to reduce the risk of a first stroke ([Ref]).

Carotid artery stenting Level of Evidence [C, G]

A randomized, controlled trial with blinded end point adjudication evaluated percutaneous carotid artery stenting versus carotid endarterectomy in patients with carotid artery stenosis. In this trial, aspirin in combination with clopidogrel was used for patients who underwent carotid artery stenting, which suggests that this antiplatelet combination is effective ([Ref]). In a prospective, single-arm multicenter trial, aspirin in combination with clopidogrel was also effective for patients who received carotid artery stenting by a transcarotid approach ([Ref]).

Based on the ASA/ACCF/AHA guideline on the management of patients with extracranial carotid and vertebral artery disease, dual antiplatelet therapy with aspirin and clopidogrel is effective at reducing ischemic complications after carotid artery stenting ([Ref]).

Colorectal cancer risk reduction, primary prevention Level of Evidence [B]

Two meta-analyses support the long-term (≥5 year) use of aspirin for primary prevention of colorectal cancer ([Ref]).

Migraine, acute treatment Level of Evidence [B, G]

In a meta-analysis of trials evaluating the treatment of acute migraine, aspirin was shown to be safe and effective ([Ref]). Based on the 2018 American Headache Society position statement on integrating new migraine treatments into clinical practice, aspirin is effective and recommended for the acute treatment of migraines of mild to moderate pain severity.

Percutaneous coronary intervention for stable ischemic heart disease Level of Evidence [G]

Based on ACC/AHA guidelines for coronary artery revascularization, aspirin in combination with a $P2Y_{12}$ inhibitor is effective for reducing ischemic complications after percutaneous coronary intervention (PCI). Aspirin is recommended for use prior to PCI and indefinitely afterwards ([Ref]).

Pericarditis, acute or recurrent Level of Evidence [C, G]

Data from a prospective, open-label clinical trial support the use of high-dose aspirin treatment in patients with acute low-risk pericarditis ([Ref]).

Based on the European Society of Cardiology guidelines for the diagnosis and management of pericardial diseases, aspirin is effective and recommended for the treatment of acute or recurrent pericarditis ([Ref]).

Peripheral artery disease Level of Evidence [G]

Based on the ACC/AHA guidelines for the management of lower extremity peripheral arterial disease (PAD), aspirin is recommended to reduce the risk of vascular events (ie, myocardial infarction, stroke, vascular death) in patients with PAD, with or without a revascularization procedure ([Ref]).

Based on the ACCP guidelines for antithrombotic therapy in PAD (9th edition), aspirin is recommended for patients with symptomatic or asymptomatic PAD to prevent cardiovascular events ([Ref]).

Polycythemia vera, prevention of thrombosis Level of Evidence [B, G]

Aspirin may be used in all patients with polycythemia vera without a history of major bleeding or gastric intolerance. Aspirin has been studied in >1,000 patients and is recommended in multiple polycythemia vera guidelines ([Ref]). Clinicians must consider the potential risk for bleeding and monitor for signs and symptoms of bleeding.

Preeclampsia prevention Level of Evidence [G]

Based on the American College of Obstetricians and Gynecologists committee opinion on low-dose aspirin use during pregnancy, low-dose aspirin is effective and recommended for use in patients at risk of preeclampsia ([Ref]).

Surgical prosthetic heart valve replacement, thromboprophylaxis Level of Evidence [G]

Based on ACC/AHA guidelines for the management of patients with valvular heart disease, in patients who receive a surgically placed bioprosthetic aortic or mitral valve and are at low bleeding risk, aspirin is not necessary immediately after surgery. In these patients, monotherapy with warfarin is recommended for 3 to 6 months, followed by discontinuation of warfarin and transition to long-term aspirin monotherapy thereafter. In patients who receive a surgically placed bioprosthetic aortic or mitral valve and are at elevated risk of bleeding, aspirin monotherapy is recommended within 24 hours after surgery and long term (ie, no initial treatment with warfarin). In patients who receive a surgically placed mechanical aortic or mitral valve, aspirin is typically not recommended; only monotherapy with warfarin is recommended, unless there is another indication for aspirin. An exception is patients who receive a surgically placed mechanical On-X aortic valve. Aspirin is recommended in combination with warfarin in these patients ([Ref]).

Transcatheter aortic valve replacement, thromboprophylaxis Level of Evidence [B, G]

Data from multiple clinical trials support the use of aspirin, with or without clopidogrel as an antithrombotic strategy for transcatheter aortic valve replacement (TAVR) ([Ref]).

Based on the ACC/AHA guideline for the management of patients with valvular heart disease, long-term aspirin is effective and recommended in patients with TAVR to reduce the risk of thromboembolic events and mortality. Clopidogrel may be used for 3 to 6 months after TAVR in addition to long-term aspirin.([Ref])

Transcatheter mitral valve repair with MitraClip device, thromboprophylaxis Level of Evidence [C]

Data from a limited number of patients in one clinical trial suggest that aspirin may be an effective component of the thromboprophylaxis regimen for transcatheter mitral valve repair when using the MitraClip device ([Ref]).

Venous thromboembolism prevention, indefinite therapy Level of Evidence [A, G]

Data from randomized, double-blind, placebo-controlled trials have demonstrated that low-dose aspirin reduces the overall risk of venous thromboembolism (VTE) recurrence in patients with a first unprovoked VTE (deep vein thrombosis [DVT] or pulmonary embolism [PE]) following completion and discontinuation of oral anticoagulant therapy ([Ref]). However, data from a double-blind, randomized, placebo-controlled trial did not demonstrate a significant reduction in the rate of VTE recurrence, although it did show a significant reduction in the rate of major vascular events ([Ref]).

Based on the ACCP guidelines for antithrombotic therapy for VTE disease, aspirin may be considered for extended treatment, after 3 months of therapeutic anticoagulation, to reduce VTE recurrence in patients who are stopping anticoagulant therapy following an unprovoked DVT or PE. Aspirin is *not* considered a reasonable alternative to anticoagulation for extended treatment because aspirin is expected to be much less effective at preventing recurrent VTE; however, in patients who have decided to stop anticoagulation (therapeutic dose or reduced intensity [eg, reduced dose apixaban or rivaroxaban]), aspirin may be considered over no aspirin therapy ([Ref]).

Venous thromboembolism prophylaxis for total hip or knee arthroplasty Level of Evidence [A]

Data from a large multicenter, randomized, double-blind, controlled trial support the use of aspirin as part of a hybrid VTE prophylaxis strategy after total hip arthroplasty (THA) or total knee arthroplasty (TKA). In this trial, patients undergoing THA or TKA received rivaroxaban for VTE prophylaxis for the first 5 postoperative days then switched to aspirin or continued rivaroxaban for an additional 30 days (35 days total) for THA or an additional 9 days (14 days total) for TKA. Aspirin was found to be noninferior to rivaroxaban at preventing symptomatic VTE, with no difference in bleeding ([Ref]). Data from another multicenter, randomized, double-blind, controlled trial support the use of aspirin as part of a hybrid VTE prophylaxis strategy after THA. Patients received dalteparin for the first 10 postoperative days and were then randomized to continue dalteparin or switch to aspirin for an additional 28 days (38 days total). Although the trial was stopped early due to slow enrollment, aspirin was found to be noninferior at preventing symptomatic VTE with no difference in bleeding ([Ref]).

Venous thromboembolism prophylaxis in lower-risk patients with multiple myeloma receiving immunomodulatory therapy Level of Evidence [B, G]

Data from a multicenter, randomized, open-label phase 3 sub-study support the use of aspirin as thromboprophylaxis in patients without a history of thromboembolism who were receiving thalidomide-based therapy for the treatment of newly diagnosed multiple myeloma ([Ref]). Data from a multicenter, randomized, open-label phase 3 sub-study support the use of aspirin as thromboprophylaxis in patients without a history of DVT or arterial thromboembolism who were receiving lenalidomide-based therapy for the treatment of newly diagnosed multiple myeloma ([Ref]). Thromboprophylaxis with aspirin was routinely used in a phase 2 study of patients with previously untreated multiple myeloma who were receiving lenalidomide and dexamethasone ([Ref]).

According to updated guidelines from the American Society of Clinical Oncology for venous thromboembolism prophylaxis and treatment in patients with cancer, aspirin is an option for pharmacologic thromboprophylaxis in lower-risk patients with multiple myeloma receiving thalidomide- or lenalidomide-based regimens ([Ref]). According to updated guidelines from the American Society of Hematology for management of venous thromboembolism: prevention and treatment in patients with cancer, low-dose aspirin is suggested for patients with multiple myeloma receiving lenalidomide, thalidomide, or pomalidomide-based regimens ([Ref]).

Level of Evidence Definitions

Level of Evidence Scale

A - Consistent evidence from well-performed randomized, controlled trials or overwhelming evidence of some other form (eg, results of the introduction of penicillin treatment) to support the off-label use. Further research is unlikely to change confidence in the estimate of benefit.

B - Evidence from randomized, controlled trials with important limitations (inconsistent results, methodological flaws, indirect or imprecise), or very strong evidence of some other research design. Further research (if performed) is likely to have an impact on confidence in the estimate of benefit and risk and may change the estimate.

C - Evidence from observational studies (eg, retrospective case series/reports providing significant impact on patient care), unsystematic clinical experience, or from potentially flawed randomized, controlled trials (eg, when limited options exist for condition). Any estimate of effect is uncertain.

G - Use has been substantiated by inclusion in at least one evidence-based or consensus-based clinical practice guideline.

Clinical Practice Guidelines

Coronary Artery Bypass Graft Surgery:

ACC/AHA, “2021 ACC/AHA Guideline for Coronary Artery Revascularization,” December 2021

AHA Scientific Statement, "Secondary Prevention After Coronary Artery Bypass Graft Surgery," February 2015

GI Bleed:

ACG/CAG, "Clinical Practice Guideline: Management of Anticoagulants and Antiplatelets During Acute Gastrointestinal Bleeding and Periendoscopic Period," March 2022

Hematology/Oncology:

BSH, "Guideline for the Diagnosis and Management of Polycythemia Vera," November 2018

ELN, "Philadelphia-Negative Classical Myeloproliferative Neoplasms: Revised Management Recommendations from European LeukemiaNet," 2018

ESMO, "Philadelphia Chromosome-Negative Chronic Myeloproliferative Neoplasms: ESMO Clinical Practice Guidelines for Diagnosis, Treatment and Follow-up," 2015

National Comprehensive Cancer Network® (NCCN), "NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)," Myeloproliferative Neoplasms

Ischemic Heart Disease:

AHA/ACC, “Guideline for the Management of Patients With Chronic Coronary Disease: a Report of the American Heart Association/American College of Cardiology Joint Committee on Clinical Practice Guidelines,” July 2023

ACC/AHA “Guideline Focused Update on Duration of Dual Antiplatelet Therapy in Patients with Coronary Artery Disease,” March 2016

AHA/ACC, "2014 AHA/ACC Guideline for the Management of Patients with Non-ST-Elevation Acute Coronary Syndromes,” September 2014.

ACCF/AHA, “2013 ACCF/AHA Guideline for the Management of ST-Elevation Myocardial Infarction,” December 2012

ESC, "Guidelines for the Management of Acute Coronary Syndromes in Patients Presenting Without Persistent ST-Segment Elevation," August 2020

Medication Safety:

“STOPP/START Criteria for Potentially Inappropriate Prescribing in Older People: Version 3,” May 2023

Pericarditis:

ESC, “2015 Guidelines for the Diagnosis and Management of Pericardial Diseases,” November 2015

Peripheral Arterial Disease:

ACC/AHA, "2024 Guideline for the Management of Lower Extremity Peripheral Artery Disease," May 2024

ACCP, “2012 Guideline on antithrombotic therapy in peripheral artery disease,” February 2012

ACCF/AHA, "2011 Focused Update of the Guideline for the Management of Patients with Peripheral Artery Disease (Updating the 2005 Guideline),” September 2011

CCS, "Canadian Cardiovascular Society 2022 Guidelines for Peripheral Arterial Disease," 2022

Preeclampsia:

USPSTF, "Aspirin Use to Prevent Preeclampsia and Related Morbidity and Mortality," 2021

Prevention:

ACC/AHA, “2019 Guideline on the primary prevention of cardiovascular disease,” March 2019

ADA, “Standards of Care in Diabetes - 2023,” January 2023

“ASCO Hereditary Colorectal Cancer Syndromes Clinical Practice Guideline [Endorsement],” December 2014

KDIGO, "KDIGO 2020 Clinical Practice Guideline for Diabetes Management in Chronic Kidney Disease," October 2020

USPSTF, "Aspirin Use to Prevent Cardiovascular Disease: US Preventive Services Task Force Recommendation Statement," April 2022

Stroke:

AHA/ASA, “Guideline for the Prevention of Stroke in Patients With Stroke and Transient Ischemic Attack,” 2021

AHA/ASA, “Guidelines for the Early Management of Patients With Acute Ischemic Stroke: 2019 Update to the 2018 Guidelines for the Early Management of Acute Ischemic Stroke,” 2019

AHA/ASA, “Guidelines for the Early Management of Patients With Acute Ischemic Stroke,” 2018

AHA/ASA, “Guidelines for the Early Management of Patients With Acute Ischemic Stroke,” February 2013

AHA/ASA, “Guidelines for the Prevention of Stroke in Women,” 2014

AHA/ASA, “Guidelines for Prevention of Stroke in Patients with Stroke and Transient Ischemic Attack,” May 2014.

AHA/ASA, “Guidelines for the Primary Prevention of Stroke,” December 2014

ASA/ACCF/AHA, "Guideline on the management of patients with extracranial carotid and vertebral artery disease," January 2011

Surgical and Procedural Management:

ACC/AHA, "2014 ACC/AHA Guideline on Perioperative Cardiovascular Evaluation and Management of Patients Undergoing Noncardiac Surgery," August 2014

ACCP, "Perioperative Management of Antithrombotic Therapy," August 2022

ESC, "Guidelines on Cardiovascular Assessment and Management of Patients Undergoing Non-Cardiac Surgery," August 2022

STS, "2012 Update to the Society of Thoracic Surgeons Guideline on Use of Antiplatelet Drugs in Patients Having Cardiac and Noncardiac Operations," November 2012

Valvular Heart Disease:

ACC/AHA, "ACC/AHA Guideline for the Management of Patients with Valvular Heart Disease," December 2020

Venous Thromboembolism:

ACCP, "Antithrombotic Therapy for VTE Disease: CHEST Guideline and Expert Panel Report," February 2016

ACCP, "Antithrombotic Therapy for VTE Disease: Second update of the CHEST Guideline and Expert Panel Report," August 2021

ASCO, "Venous Thromboembolism Prophylaxis and Treatment in Patients With Cancer: ASCO Clinical Practice Guideline Update," August 2019

ASH, "2021 Guidelines for Management of Venous Thromboembolism: Prevention and Treatment in Patients with Cancer," February 2021

ESC, "Guidelines for the diagnosis and management of acute pulmonary embolism," August 2019

Other:

AHA, "Prevention and Treatment of Thrombosis in Pediatric and Congenital Heart Disease," December 2013

Administration: IM

International product: Inject deeply into muscle.

Administration: IV

International product: May be administered IV undiluted or by IV infusion using compatible solutions (NS or D5W).

Administration: Oral

IR tablets: Administer with food or a full glass of water to minimize GI distress. In situations for which a rapid onset of action is required (eg, acute treatment of myocardial infarction), have patient chew IR tablet.

Preeclampsia (prevention): Administration as an evening dose may be more beneficial than administration in the morning ([Ref]).

ER capsules: Do not cut, crush, or chew. Administer with a full glass of water at the same time each day. Do not administer 2 hours before or 1 hour after alcohol consumption.

Bariatric surgery: Some institutions may have specific protocols that conflict with these recommendations; refer to institutional protocols as appropriate. Switch to IR formulation.

Administration: Rectal

Remove suppository from plastic packet and insert into rectum as far as possible.

Administration: Pediatric

Oral: Administer with water, food, or milk to decrease GI upset.

Immediate release:

Capsule: Swallow whole; do not cut, crush, or chew. Administer with a full glass of water.

Tablets: Do not crush or chew enteric-coated tablets; these preparations should be swallowed whole. In situations for which rapid onset of action is required, patient should chew immediate-release tablet.

Extended release: Do not cut, crush, or chew. Administer with a full glass of water at the same time each day.

Rectal: Remove suppository from plastic packet and insert into rectum as far as possible.

Storage/Stability

Store oral dosage forms (caplets, tablets, capsules) at room temperature; protect from moisture; see product-specific labeling for details. Keep suppositories in refrigerator; do not freeze. Hydrolysis of aspirin occurs upon exposure to water or moist air, resulting in salicylate and acetate, which possess a vinegar-like odor. Do not use if a strong odor is present.

Injection [International product]: Store below 25°C (77°F) and protect from light prior to reconstitution.

Preparation for Administration: Adult

Injection [International product]: After reconstitution, may dilute further with a compatible solution (eg, D5W, NS) to administer via IV infusion.

Medication Safety Issues

Sound-alike/look-alike issues:

Aspirin may be confused with Afrin

Ascriptin may be confused with Aricept

Ecotrin may be confused with Edecrin, Epogen

Halfprin may be confused with Haltran

ZORprin may be confused with Zyloprim

International issues:

Anacin 81 [Puerto Rico] may be confused with Anacin brand name for lidocaine/prilocaine [Korea]; Anacin-3 brand name for acetaminophen [Taiwan]; Anacin New brand name for acetaminophen/caffeine [India]

Cartia [multiple international markets] may be confused with Cartia XT brand name for dilTIAZem [US]

Older Adult: High-Risk Medication:

Beers Criteria: Aspirin is identified in the Beers Criteria as a potentially inappropriate medication to be avoided in patients ≥65 years of age for the primary prevention of cardiovascular disease due to increased risk for major bleeding. A potential for harm and lack of benefit when initiated as primary prevention in older adults is suggested by studies. In older adults already taking for primary prevention, deprescribing should be considered. In addition, aspirin, when used chronically at doses more than 325 mg/day, is identified in the Beers Criteria as a potentially inappropriate medication to be avoided in patients ≥65 years of age (unless alternative agents ineffective and patient can receive concomitant gastroprotective agent) due to increased risk of GI bleeding and peptic ulcer disease in older adults in high-risk category (eg, >75 years of age or receiving concomitant oral/parenteral corticosteroids, anticoagulants, or antiplatelet agents) (Beers Criteria [AGS 2023]).

Aspirin is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥65 years of age) at risk of major bleeding (eg, uncontrolled hypertension, recent significant bleeding, bleeding diathesis) or for long-term use at doses >100 mg/day. Use is also potentially inappropriate for primary prevention of cardiovascular disease, stroke prevention in patients with chronic atrial fibrillation, and long-term use for relief of symptoms in osteoarthritis or gout. Some disease states of concern include gastric antral vascular ectasia (GAVE), peptic ulcer disease, renal impairment, gastrointestinal bleeding, and severe hypertension (O’Mahony 2023).

Pediatric patients: High-risk medication:

KIDs List: Salicylates, when used in pediatric patients <18 years of age with suspicion of viral illness (influenza, chickenpox), are identified on the Key Potentially Inappropriate Drugs in Pediatrics (KIDs) list and should be used with caution due to risk of Reye syndrome (weak recommendation; very low quality of evidence) (PPA [Meyers 2020]).

Contraindications

Hypersensitivity to NSAIDs; patients with asthma, rhinitis, and nasal polyps; use in children or teenagers for viral infections, with or without fever.

Warnings/Precautions

Disease-related concerns:

- Bariatric surgery:
  - Altered absorption and efficacy: Altered absorption and efficacy may occur. The T_{max} of salicylic acid after gastric bypass was observed to be significantly shorter after surgery. C_{max} and $AUC_{0\text{-}24}$ were also significantly higher (Mitrov-Winkelmolen 2016). In a mixed surgery population (80% gastric bypass, 20% sleeve), aspirin-induced platelet reactivity was significantly reduced and correlated to the extent of weight loss after surgery (Norgard 2017).
  - Gastric ulceration: Evaluate the risk vs benefit of aspirin after surgery; if aspirin therapy is continued (eg, cardiovascular indications), use the lowest possible dose with concurrent administration of proton pump inhibitor (PPI); risk of gastric ulceration after gastric bypass and sleeve gastrectomy may be increased. A population-based study of over 20,000 patients identifying risk factors for marginal ulceration after gastric bypass suggests limited doses of aspirin may not increase risk whereas higher doses may (Sverden 2016). Another study identified that following gastric bypass, there is no significant difference in marginal ulceration rate between those given low-dose aspirin and those not on aspirin; however, long-term PPI therapy (>90 days) was found to significantly reduce marginal ulceration rate in both groups (Kang 2017).
- Bleeding disorders: Use with caution in patients with platelet and bleeding disorders.
- Dehydration: Use with caution in patients with dehydration.
- Ethanol use: Heavy ethanol use (>3 drinks/day) can increase bleeding risks.
- Gastrointestinal disease: Use with caution in patients with erosive gastritis. Avoid use in patients with active peptic ulcer disease.
- Hepatic impairment: Avoid use in severe hepatic failure.
- Renal impairment: Use high dosages (eg, analgesic or anti-inflammatory uses) with caution (NKF [Henrich 1996]; Whelton 2000). Low-dose aspirin (eg, 75 to 162 mg daily) may be safely used in patients with any degree of renal impairment (KDOQI 2005; KDOQI 2007).

Concurrent drug therapy issues:

- Thrombolytics: In the treatment of acute ischemic stroke, avoid aspirin for 24 hours following administration of a thrombolytic; administration within 24 hours increases the risk of hemorrhagic transformation (AHA/ASA [Powers 2019]; Jauch 2013).

Special populations:

- GI bleed patients: An individualized and multidisciplinary approach should be used to manage patients with an acute GI bleed who are on antiplatelet medications. Aspirin for primary prevention of cardiovascular events should be avoided in most patients with GI bleed who do not have high risk factors for cardiovascular events. However, aspirin for secondary cardiovascular prevention should *not* be discontinued in patients with established cardiovascular disease, even in the setting of a GI bleed. If held in the setting of a GI bleed, aspirin for secondary cardiovascular prevention should be resumed on the day hemostasis is confirmed by endoscopy (ACG/CAG [Abraham 2022]).

- Pediatric: When used for self-medication (OTC labeling): Children and teenagers who have or are recovering from chickenpox or flu-like symptoms should not use this product. Changes in behavior (along with nausea and vomiting) may be an early sign of Reye syndrome; patients should be instructed to contact their healthcare provider if these occur.
- Surgical patients: Patients who have recently undergone percutaneous coronary intervention with stenting or balloon angioplasty should continue antiplatelet therapy until it is safe to temporarily hold treatment. In patients undergoing CABG, aspirin may be continued until the time of surgery. Elective surgery for these patients should be delayed. Aspirin can typically be continued perioperatively in patients who undergo elective non-cardiac surgery. If aspirin interruption is necessary, discontinue therapy ≤7 days before surgery, and resume as soon as possible (eg, ≤24 hours) after surgery based on bleeding and thrombotic risks. Patient-specific situations should be discussed with cardiologist (ACC/AHA [Fleisher 2014]; ACC/AHA [Lawton 2022]; ACCP [Douketis 2022]; AHA/ACC/SCAI/ACS/ADA [Grines 2007]).

Dosage form specific issues:

- Polysorbate 80: Some dosage forms may contain polysorbate 80 (also known as Tweens). Hypersensitivity reactions, usually a delayed reaction, have been reported following exposure to pharmaceutical products containing polysorbate 80 in certain individuals (Isaksson 2002; Lucente 2000; Shelley 1995). Thrombocytopenia, ascites, pulmonary deterioration, and renal and hepatic failure have been reported in premature neonates after receiving parenteral products containing polysorbate 80 (Alade 1986; CDC 1984). See manufacturer's labeling.

Other warnings/precautions:

- Resistance: Aspirin resistance is defined as measurable, persistent platelet activation that occurs in patients prescribed a therapeutic dose of aspirin. Clinical aspirin resistance, the recurrence of some vascular event despite a regular therapeutic dose of aspirin, is considered aspirin treatment failure. Proposed mechanisms of aspirin resistance include poor adherence with therapy, poor absorption, inadequate dosage, drug interactions, increased isoprostane activity, platelet hypersensitivity to agonists, increased COX-2 activity, COX-1 polymorphism, and platelet alloantigen 2 polymorphism of platelet glycoprotein IIIa. Estimates of biochemical aspirin resistance range from 5.5% to 60% depending on the population studied and the assays used (Gasparyan 2008). Patients with aspirin resistance may have a higher risk of cardiovascular events compared to those who are aspirin sensitive (Gum 2003). Aspirin resistance is likely dose-related but may be influenced by dynamic factors yet to be identified; further research is required.

* See Cautions in AHFS Essentials for additional information.

Older Adult Considerations

Older adult patients are at high risk for adverse effects from nonsteroidal anti-inflammatory drugs (NSAIDs). Older adults can develop an asymptomatic peptic ulcer and/or hemorrhage (Savage 2005). Using the lowest effective dose for the shortest period possible is recommended. Tinnitus may be a difficult and unreliable indication of toxicity due to age-related hearing loss or eighth cranial nerve damage (Curhan 2022). CNS adverse effects such as confusion, agitation, and hallucination are generally seen in overdose or high-dose situations, but older adults may demonstrate these adverse effects at lower doses than younger adults (Savage 2005). NSAIDs seem to be a significant risk factor for falls, which can lead to morbidity and mortality in older adults. Therefore, risks and benefits should be balanced carefully in individual patients (Wongrakpanich 2018).

Low-dose aspirin increased incident anemia and decline in ferritin in otherwise healthy older adults, independent of major bleeding. Periodic monitoring of hemoglobin should be considered in older persons on aspirin (McQuilten 2023).

Warnings: Additional Pediatric Considerations

Do not use aspirin in pediatric patients <18 years of age (APS 2016) who have or who are recovering from chickenpox or flu symptoms (due to the association with Reye syndrome); when using aspirin, changes in behavior (along with nausea and vomiting) may be an early sign of Reye syndrome; instruct patients and caregivers to contact their health care provider if these symptoms occur; patients should be kept current on their influenza and varicella immunizations. Although Reye syndrome has been observed in patients receiving prolonged, high-dose aspirin therapy after Kawasaki disease presentation; it has not been observed in pediatric patients receiving low (antiplatelet) dosing regimens. Patients with Kawasaki disease and presenting with influenza or viral illness should not receive aspirin; acetaminophen is suggested as an antipyretic in these patients, and an alternate antiplatelet agent is suggested for a minimum of 2 weeks (AHA [McCrindle 2017]).

Reproductive Considerations

Low-dose aspirin has been evaluated to improve live birth rates in patients with antiphospholipid syndrome (APS) diagnosed with recurrent pregnancy loss (Hamulyák 2020; Hamulyák 2021). Recommendations for when to initiate treatment differ between guidelines. Some guidelines initiate aspirin prior to conception, while others prefer use of other agents (heparin or low-molecular-weight heparin) in patients planning to become pregnant (ACR [Sammaritano 2020]); ESHRE [Bender Atik 2018]; EULAR [Andreoli 2017]). Use of low-dose aspirin prior to conception in patients with a history of recurrent pregnancy loss but who are not diagnosed with APS has also been evaluated (Naimi 2021).

Pregnancy Considerations

Salicylate is present in umbilical cord and newborn serum following maternal use of aspirin prior to delivery (Garrettson 1975; Levy 1975; Palmisano 1969); salicylic acid and other metabolites can also be detected in the newborn urine following in utero exposure (Garrettson 1975).

Fetal outcomes are influenced by maternal dose; low dose aspirin (≤150 mg/day) is not associated with the same risks as higher doses and has a positive effect on some pregnancy outcomes (ACOG 2018; Choi 2021; Turner 2020). Adverse effects reported in the fetus following maternal use of high dose aspirin include mortality, intrauterine growth restriction, salicylate intoxication, bleeding abnormalities, and neonatal acidosis. Use of aspirin close to delivery may cause premature closure of the ductus arteriosus. Adverse effects reported in the mother include anemia, hemorrhage, prolonged gestation, and prolonged labor (Corby 1978; Østensen 1998).

Except when used in lower doses for pregnancy-related conditions, maternal use of aspirin should be avoided beginning 20 weeks gestation (FDA Safety Communication 2020).

Due to pregnancy-induced physiologic changes, some pharmacokinetic properties of aspirin may be altered (Rymark 1994; Shanmugalingam 2019).

Low-dose aspirin is recommended to prevent preeclampsia in patients at risk. The definition of risk factors varies by guideline and dosing varies by country due to dosage form availability (ACOG 2018; FIGO [Poon 2019]; SOGC [Magee 2022]; USPSTF [Davidson 2021]). The American College of Obstetricians and Gynecologists and United States Preventive Services Task Force recommend the use of low-dose aspirin for patients with ≥1 of the following high-risk factors: history of preeclampsia, multifetal gestation, chronic hypertension, type 1 or type 2 diabetes mellitus, renal disease, autoimmune disease (eg, systemic lupus erythematosus, antiphospholipid syndrome), or combinations of multiple moderate risk factors. Low-dose aspirin is also recommended for patients with ≥2 of the following moderate risk factors: nulliparity, BMI >30, family history of preeclampsia, Black race (due to social, rather than biological, factors), lower income, age ≥35 years, personal history factors (eg, low birth weight or small for GA, previous adverse pregnancy outcome, >10-year pregnancy interval), or in vitro conception. Low-dose aspirin may also be considered in patients with ≥1 of the following moderate risk factors: Black race (due to social, rather than biological, factors), lower income (ACOG 2018; ACOG 2021; USPSTF [Davidson 2021]).

In addition to preventing preeclampsia, low-dose aspirin is also recommended to improve other pregnancy outcomes and decrease the risk of thrombosis in patients with a positive antiphospholipid antibody (aPL) test. The addition of heparin or low-molecular-weight heparin may also be needed in patients with obstetric or thrombotic antiphospholipid syndrome (APS). Low-dose aspirin is also recommended for pregnant patients with systemic lupus erythematosus (SLE) (ACR [Sammaritano 2020]; EULAR [Andreoli 2017]; SMFM [Silver 2022]).

Low-dose aspirin, in addition to anticoagulation, may be used to prevent thrombosis in pregnant patients with a mechanical heart valve (ACC/AHA [Otto 2021]).

Aspirin is not recommended for the treatment of migraine headaches during pregnancy (ACOG 2022).

Breastfeeding Considerations

Salicylic acid is present in breast milk following maternal use of aspirin (Bailey 1982; Findlay 1981; Jamali 1981).

Actual breast milk concentrations of salicylic acid vary by maternal dose. In a study of six breastfeeding women, 1 to 8 months' postpartum, the highest breast milk concentration of salicylic acid was 0.0001654 mcg/mL in six women, following a maternal dose of aspirin 81 mg/day (duration of treatment not presented). Authors of this study calculated the relative infant dose (RID) of aspirin to be <1% of the weight-adjusted maternal dose (Datta 2017). In general, breastfeeding is considered acceptable when the RID is <10% (Anderson 2016; Ito 2000). Significantly higher breast milk concentrations (48.1 mcg/mL) were reported following maternal administration of aspirin 1,500 mg as a single dose (Jamali 1981).

The reported time to peak milk concentration of salicylic acid is variable (2 to 9 hours), milk concentrations decline slowly, and do not correlate strongly to maternal serum concentration (Bailey 1982; Bar-Oz 2003; Datta 2017; Findlay 1981; Jamali 1981). Salicylate is measurable in the serum (Unsworth 1987) and urine (Clark 1981) of breastfed infants following maternal use of oral aspirin. Higher salicylate concentrations may be present in breast milk following multiple maternal doses. In addition, salicylate concentrations may be higher than reported as metabolite concentrations of salicylic acid were not evaluated in most studies; the longer elimination half-life in infants compared to adults should also be considered (Bar-Oz 2003; Spigset 2000).

Metabolic acidosis was reported in a 16-day old breastfed full-term infant following maternal doses of aspirin 3.9 g/day (Clark 1981). Thrombocytopenic purpura was also reported in one infant following salicylate exposure via breast milk (maternal dose not specified) (Spigset 2000; Terragna 1967). There were no cases of diarrhea, drowsiness, or irritability noted in breastfed infants in the study which included 15 mother-infant pairs following aspirin exposure (dose, duration, and relationship to breastfeeding not provided) (Ito 1993).

Low-dose aspirin may be used in breastfeeding patients (ACOG 2022; ESC [Regitz-Zagrosek 2018]); however, standard doses of aspirin should be avoided due to the possible risk of neonatal metabolic acidosis, hemolysis, and the theoretical risk of Reye syndrome (ACOG 2022). Nonopioid analgesics are preferred for breastfeeding patients who require pain control peripartum or for surgery outside of the postpartum period. Low doses of aspirin (75 to 162 mg/day) may be used; however, other agents are preferred if higher doses are needed (ABM [Martin 2018]; ABM [Reece-Stremtan 2017]; Sachs 2013). When used for vascular indications, breastfeeding may be continued during low-dose aspirin therapy (ESC [Regitz-Zagrosek 2018]; WHO 2002). The WHO considers occasional doses of aspirin to be compatible with breastfeeding patients for the treatment of rheumatic and musculoskeletal diseases or migraine but recommends avoiding long-term therapy and consider monitoring the infant for adverse effects (hemolysis, prolonged bleeding, metabolic acidosis) (WHO 2002). High doses of aspirin in breastfeeding patients for the treatment of rheumatic and musculoskeletal diseases or migraine are not recommended (ACOG 2022; ACR [Sammaritano 2020]).

Briggs' Drugs in Pregnancy & Lactation

- Aspirin

Adverse Reactions (Significant): Considerations

GI effects

Aspirin use is associated with a 2- to 4-fold increase in upper gastrointestinal (UGI) events, such as symptomatic or complicated **gastrointestinal ulcers** and GI mucosal damage ([Ref]). Lower GI (LGI) events have also been reported with chronic aspirin use ([Ref]). Symptoms can range from mild (**dyspepsia**) to severe (peptic ulcer disease, **gastrointestinal hemorrhage**). The use of enteric-coated preparations of aspirin does not decrease the risk of UGI events ([Ref]).

Mechanism: Dose-related; related to the pharmacologic action; damage to the GI mucosa is typically caused by aspirin's effects on the epithelial and microvascular tissue of the GI tract via inhibition of cyclooxygenase-1 and subsequent prostaglandin depletion, which is gastro-protective ([Ref]). Additionally, aspirin accumulates in gastric mucosal cells, altering the permeability, resulting in an increased risk of ulceration ([Ref]).

Onset: Varied; topical gastric mucosa toxicity can be seen on endoscopy within minutes of aspirin administration ([Ref]). Other studies reported GI ulcers after at least 3 months of use ([Ref]). Risk of GI events is generally seen when beginning aspirin therapy; however, long-term administration of aspirin is associated with a greater risk of GI events ([Ref]).

Risk factors:

• Dose >100 mg ([Ref])

• Bleeding ulcers ([Ref])

- Peptic ulcer disease ([Ref])
- Concurrent use of nonsteroidal anti-inflammatory drugs ([Ref])
- Concurrent corticosteroids ([Ref])
- Concurrent dual antiplatelet and anticoagulant therapy ([Ref])
- *Helicobacter pylori* infection ([Ref])
- Smoking ([Ref])
- Excessive alcohol consumption ([Ref])
- Older age (>70 years) ([Ref])

Hypersensitivity reactions (immediate and delayed)

Hypersensitivity reactions (immediate and delayed) involving the skin (eg, **angioedema**, **urticaria**), airways (eg, dyspnea, rhinorrhea), and/or other organs ([Ref]) have been reported. Clinical phenotypes of aspirin/nonsteroidal anti-inflammatory drug (NSAID) hypersensitivity reactions include aspirin/NSAID-exacerbated respiratory disease (AERD and NERD), aspirin/NSAID-induced urticaria/angioedema (NIUA), aspirin/NSAID-exacerbated cutaneous disease (NICD) and single NSAID-induced urticaria/angioedema or **anaphylaxis** ([Ref]). Delayed hypersensitivity reactions, including **drug rash with eosinophilia and systemic symptoms** (DRESS), have also been rarely associated with aspirin ([Ref]).

Mechanism: Immediate reactions: Non–dose-related; most reactions (ie, AERD, NERD, NECD, NIUA) are nonimmunologic related to inhibition of cyclooxygenase-1 (COX-1) with subsequent activation of mast cells and eosinophils causing release of inflammatory mediators including cysteinyl-leukotrienes (cysLTs) ([Ref]). Some immediate reactions are IgE-mediated ([Ref]); although, these are rare with aspirin and more common with NSAIDs, where they can be associated with an immediate reaction specific to the specific NSAID or class of NSAID. Delayed hypersensitivity reactions, including DRESS, are T-cell-mediated ([Ref]).

Onset: Immediate hypersensitivity reactions: Rapid; occur within 1 hour of administration but may occur up to 6 hours after exposure ([Ref]). Delayed hypersensitivity reactions, including DRESS: Varied, by definition occur after 6 hours of exposure but most commonly occur 2 to 8 weeks after initiation ([Ref]).

Risk factors:

- Presence of chronic rhinosinusitis with nasal polyps, family history of AERD/NERD, and/or severe asthma may increase the risk of AERD/NERD ([Ref]). The prevalence of NERD in adult patients with asthma is ~10% to 20% ([Ref]).
- Chronic urticaria increases the risk of NECD ([Ref]). Aspirin-induced reactions are less frequent and less intense when chronic urticaria is in remission or under control ([Ref]). Approximately 12% to 30% of patients with chronic idiopathic urticaria develop exacerbations of their disease with use of aspirin and other COX-1 inhibitors ([Ref]).
- Cross-reactivity between aspirin and NSAIDs (with predominant COX-1 inhibition) have been described in patients with a history of NERD, NECD, and NIUA ([Ref]). Patients who develop a reaction (immediate or delayed) to only aspirin or a single NSAID are often able to tolerate chemically unrelated NSAIDs ([Ref]).
- Pharmacological cross-reactivity between aspirin and acetaminophen, a weak COX-1 inhibitor, and between aspirin and nonselective COX-2 inhibitors (eg, meloxicam, nimesulide) may occur ([Ref]). Due to the weak COX-1 inhibitory activity, AERD reactions associated with acetaminophen typically only occur with higher doses such as >4 g/day. Although selective COX-2 inhibitors (eg, celecoxib, etoricoxib) are generally tolerated in patients with AERD or NERD ([Ref]), pharmacological cross-reactions may occur, especially in patients with histories of urticaria/angioedema ([Ref]).
- Although earlier reports suggested that tartrazine and aspirin/NSAIDs may cross react, more recent clinical evidence has indicated that tartrazine is well-tolerated in patients with AERD/NERD ([Ref])
- Despite structural similarity between aspirin and 5-aminosalicylic acid (ASA) (mesalamine), there is lack of evidence demonstrating occurrence of AERD in association with 5-ASA ([Ref])

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. As with all drugs which may affect hemostasis, bleeding is associated with aspirin. Hemorrhage may occur at virtually any site. Risk is dependent on multiple variables including dosage, concurrent use of multiple agents which alter hemostasis, and patient susceptibility. Many adverse reactions with aspirin are dose related and are rare at low dosages. Other serious reactions are idiosyncratic, related to allergy or individual sensitivity. Accurate estimation of frequencies is not possible.

Frequency not defined:

Cardiovascular: Cardiac arrhythmia, hypotension, tachycardia

Endocrine & metabolic: Dehydration, hyperglycemia, hyperkalemia, hypoglycemia (children), increased thirst, metabolic acidosis

Gastrointestinal: Abdominal pain, dyspepsia, gastrointestinal perforation, gastrointestinal ulcer, heartburn, nausea, vomiting

Genitourinary: Postpartum hemorrhage, post-term pregnancy, prolonged labor, proteinuria, stillborn infant

Hematologic & oncologic: Disorder of hemostatic components of blood, disseminated intravascular coagulation, hemorrhage, prolonged bleeding time, prolonged prothrombin time, thrombocytopenia

Hepatic: Hepatitis, increased liver enzymes

Nervous system: Agitation, brain edema, coma, confusion, dizziness, headache, hypothermia, lethargy, seizure

Renal: Increased blood urea nitrogen, increased serum creatinine, interstitial nephritis, renal failure syndrome, renal insufficiency, renal papillary necrosis

Respiratory: Hyperventilation, laryngeal edema, pulmonary edema, respiratory alkalosis, tachypnea

Miscellaneous: Fever, low birth weight

Postmarketing:

Dermatologic: Urticaria (Kowalski 2013)

Endocrine & metabolic: Hyperuricemia (doses ≤325 mg/day) (Caspi 2000; Segal 2006)

Gastrointestinal: Gastrointestinal hemorrhage (Garcia Rodriguez 2019), pancreatitis (Hoyte 2012)

Hypersensitivity: Anaphylaxis (Kowalski 2013), angioedema (Kowalski 2013)

Immunologic: Drug reaction with eosinophilia and systemic symptoms (Kawakami 2009)

Nervous system: Intracranial hemorrhage (Huang 2019), Reye's syndrome (McGovern 2001)

Neuromuscular & skeletal: Rhabdomyolysis (Leventhal 1989)

Ophthalmic: Macular degeneration (age-related) (Li 2015)

Otic: Hearing loss (Koren 2009), tinnitus (Koren 2009)

Respiratory: Asthma (Kowalski 2013), bronchospasm (Kowalski 2013)

* See Cautions in AHFS Essentials for additional information.

Allergy and Idiosyncratic Reactions

- Salicylate Allergy/Sensitivity

Toxicology

- Salicylates

Metabolism/Transport Effects

Substrate of CYP2C9 (minor); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

Open Interactions

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Open Interactions" button above.

Miscellaneous Antiplatelets: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Abrocitinib: Aspirin may enhance the antiplatelet effect of Abrocitinib. Management: Do not use aspirin at doses greater than 81 mg/day with abrocitinib during the first 3 months of abrocitinib therapy. The abrocitinib prescribing information lists this combination as contraindicated. *Risk D: Consider therapy modification*

Acalabrutinib: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Aducanumab: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Agents with Blood Glucose Lowering Effects: Salicylates may enhance the hypoglycemic effect of Agents with Blood Glucose Lowering Effects. *Risk C: Monitor therapy*

Ajmaline: Salicylates may enhance the adverse/toxic effect of Ajmaline. Specifically, the risk for cholestasis may be increased. *Risk C: Monitor therapy*

Alcohol (Ethyl): May enhance the adverse/toxic effect of Aspirin. Specifically, alcohol may increase the bleeding risk of aspirin. Alcohol (Ethyl) may diminish the therapeutic effect of Aspirin. Specifically, alcohol may interfere with the controlled release mechanism of extended release aspirin. *Risk C: Monitor therapy*

Alendronate: Aspirin may enhance the adverse/toxic effect of Alendronate. Specifically, the incidence of upper gastrointestinal adverse events may be increased *Risk C: Monitor therapy*

Ammonium Chloride: May increase the serum concentration of Salicylates. *Risk C: Monitor therapy*

Anagrelide: May enhance the antiplatelet effect of Aspirin. *Risk C: Monitor therapy*

Angiotensin-Converting Enzyme Inhibitors: Salicylates may enhance the nephrotoxic effect of Angiotensin-Converting Enzyme Inhibitors. Salicylates may diminish the therapeutic effect of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor therapy*

Anticoagulants (Miscellaneous Agents): Aspirin may enhance the anticoagulant effect of Anticoagulants (Miscellaneous Agents). *Risk C: Monitor therapy*

Antiplatelet Agents (P2Y12 Inhibitors): Therapeutic Antiplatelets may enhance the antiplatelet effect of Antiplatelet Agents (P2Y12 Inhibitors). *Risk C: Monitor therapy*

Benzbromarone: Salicylates may diminish the therapeutic effect of Benzbromarone. *Risk C: Monitor therapy*

Calcium Channel Blockers (Nondihydropyridine): May enhance the antiplatelet effect of Aspirin. *Risk C: Monitor therapy*

Caplacizumab: May enhance the antiplatelet effect of Therapeutic Antiplatelets. Management: Avoid this combination if possible. If coadministration is required, monitor closely for bleeding. Interrupt caplacizumab if clinically significant bleeding occurs and administer von Willebrand factor concentrate to rapidly correct hemostasis, if needed. *Risk D: Consider therapy modification*

Carbonic Anhydrase Inhibitors: Salicylates may enhance the adverse/toxic effect of Carbonic Anhydrase Inhibitors. Salicylate toxicity might be enhanced by this same combination. Management: Avoid these combinations when possible.Dichlorphenamide use with high-dose aspirin as contraindicated. If another combination is used, monitor patients closely for adverse effects. Tachypnea, anorexia, lethargy, and coma have been reported. *Risk D: Consider therapy modification*

ClomiPRAMINE: May enhance the antiplatelet effect of Aspirin. *Risk C: Monitor therapy*

Collagenase (Systemic): Aspirin may enhance the adverse/toxic effect of Collagenase (Systemic). Specifically, the risk of injection site bruising and or bleeding may be increased. *Risk C: Monitor therapy*

Corticosteroids (Systemic): Salicylates may enhance the adverse/toxic effect of Corticosteroids (Systemic). These specifically include gastrointestinal ulceration and bleeding. Corticosteroids (Systemic) may decrease the serum concentration of Salicylates. Withdrawal of corticosteroids may result in salicylate toxicity. *Risk C: Monitor therapy*

Dasatinib: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Deoxycholic Acid: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Desirudin: Therapeutic Antiplatelets may enhance the anticoagulant effect of Desirudin. *Risk C: Monitor therapy*

Dexibuprofen: Aspirin may enhance the adverse/toxic effect of Dexibuprofen. Dexibuprofen may diminish the cardioprotective effect of Aspirin. *Risk X: Avoid combination*

Dexketoprofen: Salicylates may enhance the adverse/toxic effect of Dexketoprofen. Dexketoprofen may diminish the therapeutic effect of Salicylates. Salicylates may decrease the serum concentration of Dexketoprofen. Management: The use of high-dose salicylates (3 g/day or more in adults) together with dexketoprofen is inadvisable. Consider administering dexketoprofen 30-120 min after or at least 8 hrs before cardioprotective doses of aspirin to minimize any possible interaction. *Risk X: Avoid combination*

Dipyrone: May diminish the antiplatelet effect of Aspirin. Management: Use caution and consider avoiding use of dipyrone in patients treated with aspirin for the treatment or prevention of cardiovascular events or stroke. *Risk D: Consider therapy modification*

Direct Oral Anticoagulants (DOACs): Aspirin may enhance the anticoagulant effect of Direct Oral Anticoagulants (DOACs). *Risk C: Monitor therapy*

Donanemab: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Fondaparinux: Aspirin may enhance the anticoagulant effect of Fondaparinux. Management: Discontinue aspirin prior to fondaparinux therapy, if possible. If co-administration is required use caution and monitor for bleeding. *Risk D: Consider therapy modification*

Ginkgo Biloba: May enhance the anticoagulant effect of Salicylates. Management: Consider alternatives to this combination of agents. Monitor for signs and symptoms of bleeding (especially intracranial bleeding) if salicylates are used in combination with ginkgo biloba. *Risk D: Consider therapy modification*

Glycoprotein IIb/IIIa Inhibitors: Therapeutic Antiplatelets may enhance the anticoagulant effect of Glycoprotein IIb/IIIa Inhibitors. *Risk C: Monitor therapy*

Heparin: Aspirin may enhance the anticoagulant effect of Heparin. *Risk C: Monitor therapy*

Heparins (Low Molecular Weight): Aspirin may enhance the anticoagulant effect of Heparins (Low Molecular Weight). *Risk C: Monitor therapy*

Herbal Products with Anticoagulant/Antiplatelet Effects (eg, Alfalfa, Anise, Bilberry): May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Hyaluronidase: Salicylates may diminish the therapeutic effect of Hyaluronidase. *Risk C: Monitor therapy*

Ibritumomab Tiuxetan: Therapeutic Antiplatelets may enhance the antiplatelet effect of Ibritumomab Tiuxetan. *Risk C: Monitor therapy*

Ibrutinib: Therapeutic Antiplatelets may enhance the adverse/toxic effect of Ibrutinib. Specifically, the risks of bleeding and hemorrhage may be increased. *Risk C: Monitor therapy*

Icosapent Ethyl: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Influenza Virus Vaccine (Live/Attenuated): May enhance the adverse/toxic effect of Salicylates. Specifically, Reye's syndrome may develop. *Risk X: Avoid combination*

Inotersen: Therapeutic Antiplatelets may enhance the adverse/toxic effect of Inotersen. Specifically, the risk of bleeding may be increased. *Risk C: Monitor therapy*

Ketorolac (Nasal): May enhance the adverse/toxic effect of Aspirin. An increased risk of bleeding may be associated with use of this combination. Ketorolac (Nasal) may diminish the cardioprotective effect of Aspirin. Management: Concurrent use of nasal ketorolac with analgesic doses of aspirin is generally not recommended. If using low-dose, cardioprotective aspirin with nasal ketorolac, monitor the patient closely for evidence of adverse GI effects. *Risk D: Consider therapy modification*

Ketorolac (Systemic): May enhance the adverse/toxic effect of Aspirin. An increased risk of bleeding may be associated with use of this combination. Ketorolac (Systemic) may diminish the cardioprotective effect of Aspirin. *Risk X: Avoid combination*

Lecanemab: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Limaprost: May enhance the adverse/toxic effect of Therapeutic Antiplatelets. Specifically, the risk of bleeding may be increased. *Risk C: Monitor therapy*

Lipid Emulsion (Fish Oil Based): May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Loop Diuretics: Salicylates may diminish the therapeutic effect of Loop Diuretics. Loop Diuretics may increase the serum concentration of Salicylates. *Risk C: Monitor therapy*

Macimorelin: Aspirin may diminish the diagnostic effect of Macimorelin. *Risk X: Avoid combination*

Methotrexate: Salicylates may increase the serum concentration of Methotrexate. Salicylate doses used for prophylaxis of cardiovascular events are not likely to be of concern. Management: Consider avoiding coadministration of methotrexate and salicylates. If coadministration cannot be avoided, monitor for increased toxic effects of methotrexate. Salicylate doses used for prophylaxis of cardiovascular events are not likely to be of concern. *Risk D: Consider therapy modification*

Multivitamins/Fluoride (with ADE): May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Multivitamins/Minerals (with ADEK, Folate, Iron): May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Multivitamins/Minerals (with AE, No Iron): May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Nicorandil: Aspirin may enhance the adverse/toxic effect of Nicorandil. Specifically, the risk of gastrointestinal ulceration and hemorrhage may be increased. *Risk C: Monitor therapy*

Nonsteroidal Anti-Inflammatory Agents (COX-2 Selective): Aspirin may enhance the adverse/toxic effect of Nonsteroidal Anti-Inflammatory Agents (COX-2 Selective). Specifically, the risk of gastrointestinal adverse effects may be increased. Management: Concurrent use of aspirin at doses beyond cardioprotective levels is not recommended. While concurrent use of low-dose aspirin with a COX-2 inhibitor is permissible, patients should be monitored closely for signs/symptoms of GI ulceration/bleeding. *Risk D: Consider therapy modification*

Nonsteroidal Anti-Inflammatory Agents (Nonselective): May enhance the adverse/toxic effect of Salicylates. An increased risk of bleeding may be associated with use of this combination. Nonsteroidal Anti-Inflammatory Agents (Nonselective) may diminish the cardioprotective effect of Salicylates. Salicylates may decrease the serum concentration of Nonsteroidal Anti-Inflammatory Agents (Nonselective). Management: Nonselective NSAIDs may reduce aspirin's cardioprotective effects. Administer ibuprofen 30-120 minutes after immediate-release aspirin, 2 to 4 hours after extended-release aspirin, or 8 hours before aspirin. *Risk D: Consider therapy modification*

Nonsteroidal Anti-Inflammatory Agents (Topical): May enhance the adverse/toxic effect of Salicylates. Specifically, the risk of gastrointestinal (GI) toxicity is increased. Management: Coadministration of salicylates and topical NSAIDs is not recommended. If salicylates and topical NSAIDs are coadministered, ensure the benefits outweigh the risks and monitor for increased NSAID toxicities. *Risk D: Consider therapy modification*

Obinutuzumab: Therapeutic Antiplatelets may enhance the adverse/toxic effect of Obinutuzumab. Specifically, the risk of bleeding may be increased. Management: Consider avoiding coadministration of obinutuzumab and therapeutic antiplatelets, especially during the first cycle of obinutuzumab therapy. *Risk D: Consider therapy modification*

Omacetaxine: Aspirin may enhance the adverse/toxic effect of Omacetaxine. Specifically, the risk for bleeding-related events may be increased. Management: Avoid concurrent use of aspirin with omacetaxine in patients with a platelet count of less than 50,000/uL. *Risk X: Avoid combination*

Omega-3 Fatty Acids: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Pentosan Polysulfate Sodium: Therapeutic Antiplatelets may enhance the adverse/toxic effect of Pentosan Polysulfate Sodium. Specifically, the risk of hemorrhage may be increased. *Risk C: Monitor therapy*

Pirtobrutinib: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Potassium Phosphate: May increase the serum concentration of Salicylates. *Risk C: Monitor therapy*

PRALAtrexate: Salicylates may increase the serum concentration of PRALAtrexate. Salicylate doses used for prophylaxis of cardiovascular events are unlikely to be of concern. Management: Consider avoiding concomitant use of salicylates and pralatrexate. If coadministered, monitor for increased pralatrexate adverse effects. Salicylate doses used for prophylaxis of cardiovascular events are not likely to be of concern. *Risk D: Consider therapy modification*

Probenecid: Salicylates may diminish the therapeutic effect of Probenecid. *Risk X: Avoid combination*

Salicylates: May enhance the anticoagulant effect of other Salicylates. *Risk C: Monitor therapy*

Selective Serotonin Reuptake Inhibitors: May enhance the antiplatelet effect of Aspirin. *Risk C: Monitor therapy*

Selumetinib: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Serotonin/Norepinephrine Reuptake Inhibitors: May enhance the antiplatelet effect of Aspirin. *Risk C: Monitor therapy*

Sincalide: Drugs that Affect Gallbladder Function may diminish the therapeutic effect of Sincalide. Management: Consider discontinuing drugs that may affect gallbladder motility prior to the use of sincalide to stimulate gallbladder contraction. *Risk D: Consider therapy modification*

Spironolactone: Aspirin may diminish the therapeutic effect of Spironolactone. *Risk C: Monitor therapy*

Sucroferric Oxyhydroxide: May decrease the serum concentration of Aspirin. Management: Administer aspirin at least 1 hour before administration of sucroferric oxyhydroxide. *Risk D: Consider therapy modification*

Sulfinpyrazone: Salicylates may decrease the serum concentration of Sulfinpyrazone. *Risk X: Avoid combination*

Talniflumate: Aspirin may enhance the adverse/toxic effect of Talniflumate. Management: When possible, consider alternatives to this combination. Concurrent use is generally not recommended. *Risk D: Consider therapy modification*

Thiopental: Aspirin may decrease the protein binding of Thiopental. *Risk C: Monitor therapy*

Thrombolytic Agents: Therapeutic Antiplatelets may enhance the adverse/toxic effect of Thrombolytic Agents. Specifically, the risk of bleeding may be increased. *Risk C: Monitor therapy*

Ticagrelor: Aspirin may enhance the antiplatelet effect of Ticagrelor. Aspirin may diminish the therapeutic effect of Ticagrelor. More specifically, the benefits of ticagrelor relative to clopidogrel may be diminished in adult patients receiving daily aspirin doses greater than 100-150 mg daily. Management: Avoid maintenance aspirin doses greater than 150 mg/day in patients receiving ticagrelor. After any initial dose, only low-dose aspirin (75 to 100 mg/day) is recommended. *Risk D: Consider therapy modification*

Tipranavir: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Valproate Products: Salicylates may increase the serum concentration of Valproate Products. *Risk C: Monitor therapy*

Varicella Virus-Containing Vaccines: Salicylates may enhance the adverse/toxic effect of Varicella Virus-Containing Vaccines. Specifically, the risk for Reye's syndrome may increase. *Risk X: Avoid combination*

Vitamin E (Systemic): May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Vitamin K Antagonists (eg, warfarin): Aspirin may enhance the anticoagulant effect of Vitamin K Antagonists. *Risk C: Monitor therapy*

Volanesorsen: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Zanubrutinib: May enhance the antiplatelet effect of Therapeutic Antiplatelets. *Risk C: Monitor therapy*

Food Interactions

Food may decrease the rate but not the extent of oral absorption. Benedictine liqueur, prunes, raisins, tea, and gherkins have a potential to cause salicylate accumulation. Fresh fruits containing vitamin C may displace drug from binding sites, resulting in increased urinary excretion of aspirin. Curry powder, paprika, licorice; may cause salicylate accumulation. These foods contain 6 mg salicylate/100 g. An ordinary American diet contains 10-200 mg/day of salicylate. Management: Administer with food or large volume of water or milk to minimize GI upset. Limit curry powder, paprika, licorice.

Test Interactions

False-negative results for glucose oxidase urinary glucose tests (Clinistix); false-positives using the cupric sulfate method (Clinitest); also, interferes with Gerhardt test, VMA determination; 5-HIAA, xylose tolerance test and T_3 and T_4; may lead to false-positive aldosterone/renin ratio (ARR) (Funder 2016)

Genes of Interest

- CYP2C9 - Aspirin
- G6PD - Aspirin

Monitoring Parameters

Monitor for signs and symptoms of drug reaction with eosinophilia and systemic symptoms (eg, fever, rash, lymphadenopathy, eosinophilia in association with other organ system involvement such as hepatitis, nephritis, hematological abnormalities, myocarditis, myositis; early symptoms of hypersensitivity reaction may occur without rash).

Reference Range

Timing of serum samples: Peak levels usually occur 2 hours after ingestion. Salicylate serum concentrations correlate with the pharmacological actions and adverse effects observed. The serum salicylate concentration and the corresponding clinical correlations are as follows: See table.

Serum Salicylate Concentration	Desired Effects	Adverse Effects / Intoxication
~100 mcg/mL (SI: ~724 micromole/L)	Antiplatelet Antipyresis Analgesia	GI intolerance and bleeding, hypersensitivity, hemostatic defects
150 to 300 mcg/mL (SI: 1,086 to 2,172 micromole/L)	Anti-inflammatory	Mild salicylism
250 to 400 mcg/mL (SI: 1,810 to 2,896 micromole/L)	Treatment of rheumatic fever	Nausea/vomiting, hyperventilation, salicylism, flushing, sweating, thirst, headache, diarrhea, and tachycardia
>400 to 500 mcg/mL (SI: >2,896 to 3,620 micromole/L)		Respiratory alkalosis, hemorrhage, excitement, confusion, asterixis, pulmonary edema, convulsions, tetany, metabolic acidosis, fever, coma, cardiovascular collapse, renal and respiratory failure

Nursing Practice Points

Check ordered labs and report abnormalities. Monitor and instruct patients to report any signs of bleeding, signs of abdominal ulcer, or hypersensitivity reactions. Educate patient to avoid alcohol consumption within 1 hour before and 2 hours after aspirin administration. Educate pregnant patient about the correct way to perform fetal kick counts and to monitor fetal movement. Report any concerns of decreased fetal movement to physician.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Capsule, Oral:

Vazalore: 81 mg, 325 mg [contains carrageenan, fd&c blue #1 (brilliant blue), soybean oil]

Capsule Extended Release 24 Hour, Oral:

Durlaza: 162.5 mg [DSC]

Packet, Oral:

Stanback Headache Powders: 650-200-32 MG (2 ea [DSC], 6 ea [DSC], 24 ea [DSC], 50 ea [DSC])

Suppository, Rectal:

Generic: 60 mg (12 ea, 100 ea, 1000 ea); 300 mg (12 ea); 600 mg (12 ea [DSC])

Tablet, Oral:

Aspirin Adult: 325 mg [DSC] [contains corn starch]

Aspirtab Maximum Strength: 500 mg [DSC]

Bayer Aspirin: 325 mg [contains corn starch]

FT Aspirin: 325 mg [contains corn starch]

GoodSense Aspirin: 325 mg [contains corn starch]

GoodSense Aspirin: 325 mg [gluten free; contains corn starch]

Medi-First Aspirin: 325 mg [contains corn starch]

Medique Aspirin: 325 mg [contains corn starch]

Generic: 325 mg

Tablet, Oral, (buffered):

- Ascriptin: 325 mg [contains methylparaben, saccharin sodium]
- Bufferin: 325 mg [DSC]
- Bufferin: 325 mg [contains benzoic acid, fd&c blue #1 (brilliant blue)]
- Bufferin Extra Strength: 500 mg [contains benzoic acid, fd&c blue #1 (brilliant blue)]
- Tri-Buffered Aspirin: 325 mg

Tablet Chewable, Oral:

- Aspirin Childrens: 81 mg [orange flavor]
- Aspirin Childrens: 81 mg [contains fd&c yellow #6 (sunset yellow)]
- Aspirin Low Dose: 81 mg [orange flavor]
- Aspirin Low Dose: 81 mg [contains corn starch, fd&c yellow #6 (sunset yellow), saccharin sodium]
- Aspirin Low Dose: 81 mg [contains corn starch, fd&c yellow #6 (sunset yellow), saccharin sodium; orange flavor]
- Aspirin Low Dose: 81 mg [contains corn starch, fd&c yellow #6(sunset yellow)alumin lake; orange flavor]
- Bayer Low Dose: 81 mg [caffeine free, sodium free; contains corn starch, fd&c yellow #6(sunset yellow)alumin lake, saccharin sodium]
- Childrens Aspirin Low Strength: 81 mg [DSC] [caffeine free]
- FT Aspirin: 81 mg [gluten free; contains corn starch, fd&c yellow #6(sunset yellow)alumin lake, saccharin sodium; orange flavor]
- GoodSense Aspirin: 81 mg [gluten free; contains fd&c red #40(allura red ac)aluminum lake, saccharin sodium; cherry flavor]
- GoodSense Aspirin: 81 mg [gluten free; contains fd&c yellow #6(sunset yellow)alumin lake, saccharin sodium]
- GoodSense Aspirin: 81 mg [gluten free; contains fd&c yellow #6(sunset yellow)alumin lake, saccharin sodium; orange flavor]
- GoodSense Aspirin Adult Low St: 81 mg [DSC] [gluten free; contains fd&c yellow #6(sunset yellow)alumin lake, saccharin sodium; orange flavor]
- St Joseph Low Dose: 81 mg [contains corn starch, fd&c yellow #6(sunset yellow)alumin lake; orange flavor]
- Generic: 81 mg

Tablet Delayed Release, Oral:

- Aspirin 81: 81 mg [DSC] [contains fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]
- Aspirin 81: 81 mg [DSC] [contains fd&c yellow #6(sunset yellow)alumin lake, quinoline yellow (d&c yellow #10)]
- Aspirin Adult Low Dose: 81 mg [contains fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]
- Aspirin Adult Low Dose: 81 mg [DSC] [contains quinoline (d&c yellow #10) aluminum lake]
- Aspirin Adult Low Dose: 81 mg [contains quinoline (d&c yellow #10) aluminum lake, soybeans (glycine soja)]
- Aspirin Adult Low Strength: 81 mg [contains corn starch, quinoline yellow (d&c yellow #10)]
- Aspirin EC Adult Low Dose: 81 mg [contains fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]
- Aspirin EC Low Strength: 81 mg [contains corn starch, quinoline (d&c yellow #10) aluminum lake]
- Aspirin Low Dose: 81 mg [contains corn starch, fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]
- Aspirin Low Dose: 81 mg [contains fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]
- Aspirin Low Dose: 81 mg [contains fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]
- Aspirin Low Dose: 81 mg [contains quinoline (d&c yellow #10) aluminum lake]
- Bayer Aspirin: 325 mg [contains corn starch, fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]
- Bayer Aspirin EC Low Dose: 81 mg [contains corn starch, fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]

Ecotrin Maximum Strength: 500 mg [DSC]

EcPirin: 325 mg [DSC] [contains fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]

FT Aspirin Low Dose: 81 mg [contains corn starch, fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]

FT Aspirin Low Dose: 81 mg [contains fd&c red #40(allura red ac)aluminum lake, fd&c yellow #6(sunset yellow)alumin lake]

FT Aspirin Low Dose: 81 mg [contains quinoline (d&c yellow #10) aluminum lake]

FT Aspirin Low Dose: 81 mg [contains quinoline yellow (d&c yellow #10)]

FT Enteric Coated Aspirin: 325 mg [contains corn starch, fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]

GoodSense Aspirin: 325 mg [DSC] [contains corn starch, fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]

GoodSense Aspirin Low Dose: 81 mg [contains fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]

GoodSense Aspirin Low Dose: 81 mg [contains quinoline (d&c yellow #10) aluminum lake]

Miniprin Low Dose: 81 mg [DSC] [contains quinoline (d&c yellow #10) aluminum lake]

St Joseph Aspirin: 81 mg [contains fd&c red #40 (allura red ac dye), fd&c yellow #6 (sunset yellow); orange flavor]

St Joseph Low Dose: 81 mg [gluten free; contains fd&c red #40 (allura red ac dye), fd&c yellow #6 (sunset yellow); orange flavor]

Generic: 81 mg, 325 mg

Dosage Forms: International

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. **Note:** Availability of specific dosage forms may vary by region/country.

Additional dosage forms not available in the United States and/or Canada:

Solution Reconstituted, Injection: Lysine Aspirin: 900 mg (equivalent to aspirin 500 mg)

Generic Available (US)

May be product dependent

Pricing: US

Capsules (Vazalore Oral)

81 mg (per each): $0.93

325 mg (per each): $0.90

Chewable (Aspirin Oral)

81 mg (per each): $0.02 - $0.06

Chewable (St Joseph Low Dose Oral)

81 mg (per each): $0.05

Suppository (Aspirin Rectal)

300 mg (per each): $1.46

Tablet, EC (Aspirin Oral)

81 mg (per each): $0.01 - $0.03

325 mg (per each): $0.02 - $0.05

Tablet, EC (Bayer Aspirin EC Low Dose Oral)

81 mg (per each): $0.04

Tablet, EC (Bayer Aspirin Oral)

325 mg (per each): $0.08

Tablets (Ascriptin Oral)

325 mg (per each): $0.08

Tablets (Aspirin Oral)

325 mg (per each): $0.01 - $0.03

Tablets (Bayer Aspirin Oral)

325 mg (per each): $0.14

Tablets (Bufferin Extra Strength Oral)

500 mg (per each): $0.10

Tablets (Bufferin Oral)

325 mg (per each): $0.07

Tablets (Medi-First Aspirin Oral)

325 mg (per each): $0.04

Tablets (Medique Aspirin Oral)

325 mg (per each): $0.04

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Mechanism of Action

Irreversibly inhibits cyclooxygenase-1 and 2 (COX-1 and 2) enzymes, via acetylation, which results in decreased formation of prostaglandin precursors; irreversibly inhibits formation of prostaglandin derivative, thromboxane A_2, via acetylation of platelet cyclooxygenase, thus inhibiting platelet aggregation; has antipyretic, analgesic, and anti-inflammatory properties

Pharmacokinetics (Adult Data Unless Noted)

Onset: Immediate release: Platelet inhibition: Nonenteric-coated: <1 hour; enteric-coated: 3 to 4 hours (Eikelboom 2012). **Note:** Chewing nonenteric-coated or enteric-coated tablets results in inhibition of platelet aggregation within 20 minutes (Eikelboom 2012; Feldman 1999; Sai 2011).

Duration: Immediate release: 4 to 6 hours; however, platelet inhibitory effects last the lifetime of the platelet (~10 days) due to its irreversible inhibition of platelet COX-1 (Eikelboom 2012).

Absorption: Immediate release: Rapidly absorbed in stomach and upper intestine (Eikelboom 2012); Extended-release capsule: Rate of absorption is dependent upon food, alcohol, and gastric pH.

Distribution: V_d: 10 L; readily into most body fluids and tissues; hydrolyzed to salicylate (active) by esterases in the GI mucosa, red blood cells, synovial fluid and blood

Protein binding: Concentration dependent; as salicylate concentration increases, protein binding decreases: ~90% to 94% (to albumin) at concentrations ≤80 mcg/mL (Rosenberg 1981; Juurlink 2015); ~30% with concentrations seen in overdose (Juurlink 2015).

Metabolism: Hydrolyzed to salicylate (active) by esterases in GI mucosa, red blood cells, synovial fluid, and blood; metabolism of salicylate occurs primarily by hepatic conjugation; metabolic pathways are saturable

Bioavailability: Immediate release: 50% to 75% reaches systemic circulation

Half-life elimination: Parent drug: Plasma concentration: 15 to 20 minutes; Salicylates (dose dependent): 3 hours at lower doses (300 to 600 mg), 5 to 6 hours (after 1 g), 10 hours with higher doses

Time to peak, serum: Immediate release: ~1 to 2 hours (nonenteric-coated), 3 to 4 hours (enteric-coated) (Eikelboom 2012); Extended-release capsule: ~2 hours. **Note:** Chewing nonenteric-coated tablets results in a time to peak concentration of 20 minutes (Feldman 1999). Chewing enteric-coated tablets results in a time to peak concentration of 2 hours (Sai 2011).

Excretion: Urine (75% as salicyluric acid, 10% as salicylic acid)

Dental Use

Treatment of postoperative pain

* See Uses in AHFS Essentials for additional information.

Dental: Local Anesthetic/Vasoconstrictor Precautions

No information available to require special precautions

Dental Health Professional Considerations

The Food and Drug Administration (FDA), has issued a letter updating information and considerations regarding the use of ibuprofen (400 mg doses) in patients who are taking low dose aspirin (81 mg, immediate release; not enteric coated) for cardioprotection and stroke prevention. Ibuprofen, at these doses, may interfere with aspirin's antiplatelet effect depending upon when it is administered. Patients initiated on aspirin first (for ~1 week) then ibuprofen (400 mg 3 times/day for 10 days) seem to maintain aspirin's platelet effect (Cryer, 2005). Ibuprofen has the greatest impact on aspirin if administered less than 8 hours before aspirin (Catella-Lawson, 2001).

Patients may require counseling about the appropriate timing of ibuprofen dosing in relationship to aspirin therapy. With occasional use of ibuprofen, a clinically-significant interaction with aspirin in unlikely. To avoid interference during chronic dosing, a single dose of ibuprofen should be taken 30 to 120 minutes after aspirin ingestion or at least 8 hours should elapse after ibuprofen dosing before giving aspirin (Catella-Lawson, 2001; FDA, 2006).

The clinical implications of the interaction are unclear. There have not been any clinical endpoint studies conducted at this time. Avoidance of this interaction is potentially important because aspirin's vascular protection could be decreased or negated.

Other nonselective NSAIDs may have potential for a similar interaction with aspirin. Such has been described with naproxen (Capone, 2005). Acetaminophen does not appear to interfere with the antiplatelet effect of aspirin. Other clinical scenarios (use of smaller ibuprofen doses, other aspirin products, other doses of aspirin) have not been evaluated.

Additional information is available at:
http://www.fda.gov/Drugs/DrugSafety/PostmarketDrugSafetyInformationforPatientsandProviders/ucm125222.htm

Dental: Effects on Dental Treatment

Key adverse event(s) related to dental treatment: As with all drugs which may affect hemostasis, bleeding is associated with aspirin. Hemorrhage may occur at virtually any site; risk is dependent on multiple variables including dosage, concurrent use of multiple agents which alter hemostasis, and patient susceptibility. Many adverse effects of aspirin are dose related, and are rare at low dosages. Other serious reactions are idiosyncratic, related to allergy or individual sensitivity (see Dental Health Professional Considerations).

Aspirin as sole antiplatelet agent: Patients taking aspirin for ischemic stroke prevention are safe to continue it during dental procedures (Armstrong, 2013).

Concurrent aspirin use with other antiplatelet agents: Aspirin in combination with clopidogrel (Plavix), prasugrel (Effient), or ticagrelor (Brilinta) is the primary prevention strategy against stent thrombosis after placement of drug-eluting metal stents in coronary patients. Premature discontinuation of combination antiplatelet therapy (ie, dual antiplatelet therapy) strongly increases the risk of a catastrophic event of stent thrombosis leading to myocardial infarction and/or death, so says a science advisory issued in January 2007 from the American Heart Association in collaboration with the American Dental Association and other professional healthcare organizations. The advisory stresses a 12-month therapy of dual antiplatelet therapy after placement of a drug-eluting stent in order to prevent thrombosis at the stent site. Any elective surgery should be postponed for 1 year after stent implantation, and if surgery must be performed, consideration should be given to continuing the antiplatelet therapy during the perioperative period in high-risk patients with drug-eluting stents.

This advisory was issued from a science panel made up of representatives from the American Heart Association (AHA), the American College of Cardiology, the Society for Cardiovascular Angiography and Interventions, the American College of Surgeons, the American Dental Association (ADA), and the American College of Physicians (Grines, 2007).

Dental: Effects on Bleeding

Aspirin irreversibly inhibits platelet aggregation which can prolong bleeding. Upon discontinuation, normal platelet function returns only when new platelets are released (~7 to 10 days). However, in the case of dental surgery, there is no scientific evidence to support discontinuation of aspirin. This was recently supported by the American Academy of Neurology in patients with ischemic cerebrovascular disease (Armstrong, 2013). A recent study compared blood loss after a single tooth extraction in coronary artery disease patients who were either on aspirin (100 mg daily) or off aspirin for the extraction. The mean volume of bleeding was not statistically different between the groups. Local hemostatic measures were sufficient to control bleeding and there were no reported episodes of hemorrhaging intra- or postoperatively (Medeiros, 2011).

Dental Usual Dosing

Postoperative pain:

Analgesic and antipyretic: Oral, rectal:

Children: 10 to 15 mg/kg/dose every 4 to 6 hours, up to a total of 4 g/day

Adults: 325 to 650 mg every 4 to 6 hours up to 4 g/day

Anti-inflammatory: Oral: Initial:

Children: 60 to 90 mg/kg/day in divided doses; usual maintenance: 80 to 100 mg/kg/day divided every 6 to 8 hours; monitor serum concentrations

Adults: 2.4 to 3.6 g/day in divided doses; usual maintenance: 3.6 to 5.4 g/day; monitor serum concentrations

Related Information

- Beers Criteria 2023: Potentially Inappropriate Medication Use in Older Adults ≥65 Years of Age
- KIDs List
- Oral Antiplatelet Comparison Chart
- Oral Medications That Should Not Be Crushed or Altered

Index Terms

Acetylsalicylic Acid; ASA; Baby Aspirin; DL-Lysine Acetylsalicylate; Lysine Aspirin

References

2023 American Geriatrics Society Beers Criteria Update Expert Panel. American Geriatrics Society 2023 updated AGS Beers Criteria for potentially inappropriate medication use in older adults. *J Am Geriatr Soc*. 2023;71(7):2052-2081. doi:10.1111/jgs.18372[PubMed 37139824]

Abraham NS, Barkun AN, Sauer BG, et al. American College of Gastroenterology-Canadian Association of Gastroenterology clinical practice guideline: management of anticoagulants and antiplatelets during acute gastrointestinal bleeding and the periendoscopic period. *J Can Assoc Gastroenterol.* 2022;5(2):100-101. doi:10.1093/jcag/gwac010[PubMed 35368325]

Abramson SB. Aspirin: Mechanism of action, major toxicities, and use in rheumatic diseases. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed September 30, 2021.

Adams DH, Popma JJ, Reardon MJ, et al; US CoreValve Clinical Investigators. Transcatheter aortic-valve replacement with a self-expanding prosthesis. *N Engl J Med*. 2014;370(19):1790-1798. doi:10.1056/NEJMoa1400590[PubMed 24678937]

Adler Y, Charron P, Imazio M, et al; ESC Scientific Document Group. 2015 ESC guidelines for the diagnosis and management of pericardial diseases: The Task Force for the Diagnosis and Management of Pericardial Diseases of the European Society of Cardiology (ESC) endorsed by: The European Association for Cardio-Thoracic Surgery (EACTS). *Eur Heart J*. 2015;36(42):2921-2964. doi:10.1093/eurheartj/ehv318[PubMed 26320112]

Alade SL, Brown RE, Paquet A. Polysorbate 80 and E-Ferol toxicity. *Pediatrics.* 1986;77(4):593-597.[PubMed 3960626]

Alonso-Coello P, Bellmunt S, McGorrian C, et al. Antithrombotic therapy in peripheral artery disease: Antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians evidence-based clinical practice guidelines. *Chest*. 2012;141(2)(suppl):e669S-e690S. doi:10.1378/chest.11-2307[PubMed 22315275]

American Academy of Pediatrics (AAP). In: Kimberlin DW, Barnett ED, Lynfield R, Sawyer MH, eds. *Red Book: 2021 Report of the Committee on Infectious Diseases*. 32nd ed. American Academy of Pediatrics; 2021.

American College of Obstetricians and Gynecologists (ACOG). ACOG Committee Opinion No. 743: Low-dose aspirin use during pregnancy. *Obstet Gynecol*. 2018;132(1):e44-e52. doi:10.1097/AOG.0000000000002708[PubMed 29939940]

American College of Obstetricians and Gynecologists (ACOG). Headaches in pregnancy and postpartum: ACOG clinical practice guideline no. 3. *Obstet Gynecol*. 2022;139(5):944-972. doi:10.1097/AOG.0000000000004766[PubMed 35576364]

American College of Obstetricians and Gynecologists (ACOG) Practice Advisory. Low-dose aspirin use for the prevention of preeclampsia and related morbidity and mortality. 2021. https://www.acog.org/clinical/clinical-guidance/practice-advisory/articles/2021/12/low-dose-aspirin-use-for-the-prevention-of-preeclampsia-and-related-morbidity-and-mortality. Accessed November 21, 2022.

American Diabetes Association (ADA). Standards of care in diabetes–2023. *Diabetes Care*. 2023;46(suppl 1):S1-S291. https://diabetesjournals.org/care/issue/46/Supplement_1. Accessed January 4, 2023.

American Headache Society. The American Headache Society position statement on integrating new migraine treatments into clinical practice [published correction appears in *Headache*. 2019;59(4):650-651]. *Headache*. 2019;59(1):1-18. doi:10.1111/head.13456[PubMed 30536394]

American Pain Society (APS). *Principles of Analgesic Use in the Treatment of Acute Pain and Cancer Pain*. 7th ed. Glenview, IL: 2016.

Anderson DR, Dunbar MJ, Bohm ER, et al. Aspirin versus low-molecular-weight heparin for extended venous thromboembolism prophylaxis after total hip arthroplasty: a randomized trial. *Ann Intern Med*. 2013;158(11):800-806. doi:10.7326/0003-4819-158-11-201306040-00004[PubMed 23732713]

Anderson DR, Dunbar M, Murnaghan J, et al. Aspirin or rivaroxaban for VTE prophylaxis after hip or knee arthroplasty. *N Engl J Med*. 2018;378(8):699-707. doi:10.1056/NEJMoa1712746[PubMed 29466159]

Anderson PO, Sauberan JB. Modeling drug passage into human milk. *Clin Pharmacol Ther*. 2016;100(1):42-52.[PubMed 27060684]

Andreoli L, Bertsias GK, Agmon-Levin N, et al. EULAR recommendations for women's health and the management of family planning, assisted reproduction, pregnancy and menopause in patients with systemic lupus erythematosus and/or antiphospholipid syndrome. *Ann Rheum Dis*. 2017;76(3):476-485. doi:10.1136/annrheumdis-2016-209770[PubMed 27457513]

Aranki S, Maltais S, Toeg HD. Medical therapy to prevent complications after coronary artery bypass graft surgery. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed March 8, 2021.

Armstrong MJ, Gronseth G, Anderson DC, et al. Summary of evidence-based guideline: Periprocedural management of antithrombotic medications in patients with ischemic cerebrovascular disease: Report of the Guideline Development Subcommittee of the American Academy of Neurology. *Neurology*. 2013;80(22):2065-2069.[PubMed 23713086]

Arnett DK, Blumenthal RS, Albert MA, et al. 2019 ACC/AHA guideline on the primary prevention of cardiovascular disease: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines [published correction appears in *Circulation*. 2019;140(11):e649-e650]. *Circulation*. 2019;140(11):e596-e646. doi:10.1161/CIR.0000000000000678[PubMed 30879355]

Aronoff GR, Bennett WM, Berns JS, et al. *Drug Prescribing in Renal Failure: Dosing Guidelines for Adults and Children, 5th ed*. Philadelphia, PA: American College of Physicians; 2007;20-136.

Aspirin [prescribing information]. Morristown, NJ: Bayer Corporation.

Ayala DE, Ucieda R, Hermida RC. Chronotherapy with low-dose aspirin for prevention of complications in pregnancy. *Chronobiol Int.* 2013;30(1-2):260-279. doi:10.3109/07420528.2012.717455[PubMed 23004922]

Bailey DN, Weibert RT, Naylor AJ, Shaw RF. A study of salicylate and caffeine excretion in the breast milk of two nursing mothers. *J Anal Toxicol*. 1982;6(2):64-68.[PubMed 7098450]

Bar-Oz B, Bulkowstein M, Benyamini L, et al. Use of antibiotic and analgesic drugs during lactation. *Drug Saf*. 2003;26(13):925-935. doi:10.2165/00002018-200326130-00002[PubMed 14583068]

Barbui T, Finazzi G. Evidence-based management of polycythemia vera. *Best Pract Res Clin Haematol.* 2006;19(3):483-493. doi:10.1016/j.beha.2005.07.007[PubMed 16781485]

Barbui T, Tefferi A, Vannucchi AM, et al. Philadelphia chromosome-negative classical myeloproliferative neoplasms: revised management recommendations from European LeukemiaNet. *Leukemia*. 2018;32(5):1057-1069. doi:10.1038/s41375-018-0077-1[PubMed 29515238]

Bayer Chewable Low Dose Aspirin 81 mg (aspirin) [prescribing information]. Whippany, NJ: Bayer Corporation; December 2019.

Bayer Regular Strength Aspirin 325 mg Safety Coated Caplets (aspirin) [prescribing information]. Whippany, NJ: Bayer Corporation; March 2018.

Bayer Safety Coated 81 mg (Regimen) (aspirin) [prescribing information]. Whippany, NJ: Bayer Corporation; February 2021.

Becattini C, Agnelli G, Schenone A, et al; WARFASA Investigators. Aspirin for preventing the recurrence of venous thromboembolism [published correction appears in *N Engl J Med*. 2012;367(16):1573]. *N Engl J Med*. 2012;366(21):1959-1967. doi:10.1056/NEJMoa1114238[PubMed 22621626]

Bender Atik R, Christiansen OB, Elson J, et al; ESHRE Guideline Group on RPL. ESHRE guideline: recurrent pregnancy loss. *Hum Reprod Open*. 2018;2018(2):hoy004. doi:10.1093/hropen/hoy004[PubMed 31486805]

Blanca-Lopez N, Soriano V, Garcia-Martin E, Canto G, Blanca M. NSAID-induced reactions: classification, prevalence, impact, and management strategies. *J Asthma Allergy*. 2019;12:217-233. doi:10.2147/JAA.S164806[PubMed 31496752]

Blumenthal KG, Peter JG, Trubiano JA, Phillips EJ. Antibiotic allergy. *Lancet*. 2019;393(10167):183-198. doi:10.1016/S0140-6736(18)32218-9[PubMed 30558872]

Bonaca MP, Bauersachs RM, Anand SS, et al. Rivaroxaban in peripheral artery disease after revascularization. *N Engl J Med*. 2020;382(21):1994-2004. doi:10.1056/NEJMoa2000052[PubMed 32222135]

Bonaca MP, Bhatt DL, Cohen M, et al; PEGASUS-TIMI 54 Steering Committee and Investigators. Long-term use of ticagrelor in patients with prior myocardial infarction. *N Engl J Med*. 2015;372(19):1791-1800. doi:10.1056/NEJMoa1500857[PubMed 25773268]

Bradley LM, Midgley FM, Watson DC, et al. Anticoagulation Therapy in Children With Mechanical Prosthetic Cardiac Valves. *Am J Cardiol*. 1985;56(8):533-535.[PubMed 4036839]

Brighton TA, Eikelboom JW, Mann K, et al; ASPIRE Investigators. Low-dose aspirin for preventing recurrent venous thromboembolism. *N Engl J Med*. 2012;367(21):1979-1987. doi:10.1056/NEJMoa1210384[PubMed 23121403]

Brilinta (ticagrelor) [prescribing information]. Wilmington, DE: AstraZeneca Pharmaceuticals LP; August 2021.

Brockow K, Przybilla B, Aberer W, et al. Guideline for the diagnosis of drug hypersensitivity reactions: S2K-Guideline of the German Society for Allergology and Clinical Immunology (DGAKI) and the German Dermatological Society (DDG) in collaboration with the Association of German Allergologists (AeDA), the German Society for Pediatric Allergology and Environmental Medicine (GPA), the German Contact Dermatitis Research Group (DKG), the Swiss Society for Allergy and Immunology (SGAI), the Austrian Society for Allergology and Immunology (ÖGAI), the German Academy of Allergology and Environmental Medicine (DAAU), the German Center for Documentation of Severe Skin Reactions and the German Federal Institute for Drugs and Medical Products (BfArM). *Allergo J Int*. 2015;24(3):94-105. doi:10.1007/s40629-015-0052-6[PubMed 26120552]

Brott TG, Halperin JL, Abbara S, et al; American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines; American Stroke Association; American Association of Neuroscience Nurses; American Association of Neurological Surgeons; American College of Radiology; American Society of Neuroradiology; Congress of Neurological Surgeons; Society of Atherosclerosis Imaging and Prevention; Society for Cardiovascular Angiography and Interventions; Society of Interventional Radiology; Society of NeuroInterventional Surgery; Society for Vascular Medicine; Society for Vascular Surgery; American Academy of Neurology and Society of Cardiovascular Computed Tomography. 2011 ASA/ACCF/AHA/AANN/AANS/ACR/ASNR/CNS/SAIP/SCAI/SIR/SNIS/SVM/SVS guideline on the management of patients with extracranial carotid and vertebral artery disease. *Stroke*. 2011;42(8):e464-e540. doi:10.1161/STR.0b013e3182112cc2[PubMed 21282493]

Brott TG, Hobson RW 2nd, Howard G, et al; CREST Investigators. Stenting versus endarterectomy for treatment of carotid-artery stenosis [published corrections appear in *N Engl J Med*. 2010;363(2):198; *N Engl J Med*. 2010;363(5):498]. *N Engl J Med*. 2010;363(1):11-23. doi:10.1056/NEJMoa0912321[PubMed 20505173]

Brouwer J, Nijenhuis VJ, Delewi R, et al. Aspirin with or without clopidogrel after transcatheter aortic-valve implantation. *N Engl J Med*. 2020;383(15):1447-1457. doi:10.1056/NEJMoa2017815[PubMed 32865376]

Capone ML, Sciulli MG, Tacconelli S, et al. Pharmacodynamic interaction of naproxen with low-dose aspirin in healthy subjects. *J Am Coll Cardiol*. 2005;45(8):1295-1301. doi:10.1016/j.jacc.2005.01.045[PubMed 15837265]

Carapetis J, Brown A, Maguire G, Walsh W. *The Australian guideline for prevention, diagnosis and management of acute rheumatic fever and rheumatic heart disease*. 2nd ed. Casuarina, NT, Australia: RHD Australia, Menzies School of Health Research; 2012.

Caspi D, Lubart E, Graff E, Habot B, Yaron M, Segal R. The effect of mini-dose aspirin on renal function and uric acid handling in elderly patients. *Arthritis Rheum*. 2000;43(1):103-108. doi:10.1002/1529-0131(200001)43:1<103::AID-ANR13>3.0.CO;2-C[PubMed 10643705]

Catella-Lawson F, Reilly MP, Kapoor SC, et al. Cyclooxygenase inhibitors and the antiplatelet effects of aspirin. *N Engl J Med*. 2001;345(25):1809-1817. doi:10.1056/NEJMoa003199[PubMed 11752357]

Cea Soriano L, Lanas A, Soriano-Gabarró M, García Rodríguez LA. Incidence of upper and lower gastrointestinal bleeding in new users of low-dose aspirin. *Clin Gastroenterol Hepatol*. 2019;17(5):887-895.e6. doi:10.1016/j.cgh.2018.05.061[PubMed 29908361]

Centers for Disease Control (CDC). Unusual syndrome with fatalities among premature infants: association with a new intravenous vitamin E product. *MMWR Morb Mortal Wkly Rep*. 1984;33(14):198-199. http://www.cdc.gov/mmwr/preview/mmwrhtml/00000319.htm[PubMed 6423951]

Chan A. NSAIDs (including aspirin): role in prevention of colorectal cancer. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed May 4, 2020.

Chimowitz MI, Lynn MJ, Derdeyn CP, et al; SAMMPRIS Trial Investigators. Stenting versus aggressive medical therapy for intracranial arterial stenosis. *N Engl J Med*. 2011;365(11):993-1003. doi:10.1056/NEJMoa1105335[PubMed 21899409]

Choi YJ, Shin S. Aspirin prophylaxis during pregnancy: a systematic review and meta-analysis. *Am J Prev Med*. 2021;61(1):e31-e45. doi:10.1016/j.amepre.2021.01.032[PubMed 33795180]

Clark JH, Wilson WG. A 16-day-old breast-fed infant with metabolic acidosis caused by salicylate. *Clin Pediatr (Phila)*. 1981;20(1):53-54.[PubMed 7449246]

Corby DG. Aspirin in pregnancy: maternal and fetal effects. *Pediatrics*. 1978;62(2)(suppl):930-937.[PubMed 364401]

Cryer B, Mahaffey KW. Gastrointestinal ulcers, role of aspirin, and clinical outcomes: pathobiology, diagnosis, and treatment. *J Multidiscip Healthc*. 2014;7:137-146. doi:10.2147/JMDH.S54324[PubMed 24741318]

Cryer B, Verlin RG, Cooper SA, et al, “Double-Blind, Randomized, Parallel, Placebo-Controlled Study of Ibuprofen Effects on Thromboxane B2 Concentrations in Aspirin-Treated Healthy Adult Volunteers,” *Clin Ther*, 2005, 27(2):185-191.[PubMed 15811481]

Cucchiara BL, Messé SR. Long-term antithrombotic therapy for the secondary prevention of ischemic stroke. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed November 10, 2022.

Curhan SG, Glicksman J, Wang M, Eavey RD, Curhan GC. Longitudinal study of analgesic use and risk of incident persistent tinnitus. *J Gen Intern Med*. 2022;37(14):3653-3662. doi:10.1007/s11606-021-07349-5[PubMed 35132561]

Datta P, Rewers-Felkins K, Kallem RR, Baker T, Hale TW. Transfer of low dose aspirin into human milk. *J Hum Lact*. 2017;33(2):296-299.[PubMed 28418802]

Davidson KW, Barry MJ, Mangione CM, et al; US Preventive Services Task Force (USPSTF). Aspirin use to prevent preeclampsia and related morbidity and mortality: US Preventive Services Task Force recommendation statement. *JAMA*. 2021;326(12):1186-1191. doi:10.1001/jama.2021.14781[PubMed 34581729]

Department of Health and Human Services, Food and Drug Administration. Labeling for Oral and Rectal Over-the-Counter Drug Products Containing Aspirin and Nonaspirin Salicylates; Reye's Syndrome Warning; Final Rule, (29 CFR Part 201). *Fed Regist*. April 17, 2003: 18861-18869.[PubMed 10177739]

Doña I, Blanca-López N, Jagemann LR, et al. Response to a selective COX-2 inhibitor in patients with urticaria/angioedema induced by nonsteroidal anti-inflammatory drugs. *Allergy*. 2011;66(11):1428-1433. doi:10.1111/j.1398-9995.2011.02684.x[PubMed 21834936]

Douketis JD. Prevention of venous thromboembolism in adults undergoing hip fracture repair or hip or knee replacement. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed March 16, 2023.

Douketis JD, Spyropoulos AC, Murad MH, et al. Perioperative management of antithrombotic therapy: an American College of Chest Physicians clinical practice guideline. *Chest*. 2022;162(5):e207-e243. doi:10.1016/j.chest.2022.07.025[PubMed 35964704]

Durlaza (aspirin) [prescribing information]. North Haven, CT: New Haven Pharmaceuticals; September 2015.

Eikelboom JW, Hirsh J, Spencer FA, Baglin TP, Weitz JI. Antiplatelet drugs: antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians evidence-based clinical practice guidelines. *Chest*. 2012:141(2)(suppl):e89S-e119S. [PubMed 22315278]

el Makhlouf A, Friedli B, Oberhänsli I, et al. Prosthetic Heart Valve Replacement in Children: Results and Follow-up of 273 Patients. *J Thorac Cardiovasc Surg*. 1987;93(1):80-85.[PubMed 3796033]

Expert opinion. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Fairman RM. Carotid endarterectomy. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed March 10, 2021.

Feldman M, Cryer B. Aspirin absorption rates and platelet inhibition times with 325-mg buffered aspirin tablets (chewed or swallowed intact) and with buffered aspirin solution. *Am J Cardiol*. 1999;84(4):404-409.[PubMed 10468077]

Fernandez JS, Sadaniantz A. Medical and revascularization management in acute coronary syndrome in renal patients. *Semin Nephrol*. 2001;21(1):25-35.[PubMed 11172556]

Findlay JW, DeAngelis RL, Kearney MF, Welch RM, Findlay JM. Analgesic drugs in breast milk and plasma. *Clin Pharmacol Ther*. 1981;29(5):625-633.[PubMed 7214793]

Fleisher LA, Fleischmann KE, Auerbach AD, et al. 2014 ACC/AHA guideline on perioperative cardiovascular evaluation and management of patients undergoing noncardiac surgery: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;130(24):2215-2245. doi:10.1161/CIR.0000000000000105[PubMed 25085962]

Funder JW, Carey RM, Mantero F, et al. The management of primary aldosteronism: case detection, diagnosis, and treatment: an Endocrine Society Clinical Practice Guideline. *J Clin Endocrinol Metab*. 2016;101(5):1889-1916. doi:10.1210/jc.2015-4061[PubMed 26934393]

Furie KL, Kasner SE, Adams RJ, et al; American Heart Association Stroke Council, Council on Cardiovascular Nursing, Council on Clinical Cardiology, and Interdisciplinary Council on Quality of Care and Outcomes Research. Guidelines for the prevention of stroke in patients with stroke or transient ischemic attack: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke.* 2011;42(1):227-276.[PubMed 20966421]

García Rodríguez LA, Lanas A, Soriano-Gabarró M, Cea Soriano L. Low-dose aspirin and risk of upper/lower gastrointestinal bleeding by bleed severity: a cohort study with nested case-control analysis using primary care electronic health records from the United Kingdom. *Ann Med*. 2019;51(2):182-192. doi:10.1080/07853890.2019.1591635[PubMed 31025592]

Garrettson LK, Procknal JA, Levy G. Fetal acquisition and neonatal elimination of a large amount of salicylate. Study of a neonate whose mother regularly took therapeutic doses of aspirin during pregnancy. *Clin Pharmacol Ther*. 1975;17(1):98-103.[PubMed 235394]

Gasparyan AY, Watson T, Lip GYH. The role of aspirin in cardiovascular prevention; implications of aspirin resistance. *Am J Coll Cardiol*. 2008;51(9):1829-1843.[PubMed 18466797]

Gerhard-Herman MD, Gornik HL, Barrett C, et al. 2016 AHA/ACC guideline on the management of patients with lower extremity peripheral artery disease: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2017;69(11):e71-e126. doi:10.1016/j.jacc.2016.11.007[PubMed 27851992]

Giancane G, Consolaro A, Lanni S, Davì S, Schiappapietra B, Ravelli A. Juvenile idiopathic arthritis: diagnosis and treatment. *Rheumatol Ther*. 2016;3(2):187-207. doi:10.1007/s40744-016-0040-4[PubMed 27747582]

Giglia TM, Massicotte MP, Tweddell JS, et al. Prevention and treatment of thrombosis in pediatric and congenital heart disease: a scientific statement from the American Heart Association. *Circulation*. 2013;128(24):2622-2703.[PubMed 24226806]

Gorelik M, Chung SA, Ardalan K, et al. 2021 American College of Rheumatology/Vasculitis Foundation guideline for the management of Kawasaki disease. *Arthritis Care Res (Hoboken)*. 2022;74(4):538-548. doi:10.1002/acr.24838[PubMed 35257507]

Gornik HL, Aronow HD, Goodney PP, et al. 2024 ACC/AHA/AACVPR/APMA/ABC/SCAI/SVM/SVN/SVS/SIR/VESS guideline for the management of lower extremity peripheral artery disease: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2024;149(24):e1313-e1410. doi:10.1161/CIR.0000000000001251[PubMed 38743805]

Grines CL, Bonow RO, Casey DE, et al. AHA/ACC/SCAI/ACS/ADA Science Advisory, Prevention of Premature Discontinuation of Dual Antiplatelet Therapy in Patients With Coronary Artery Stents. A Science Advisory From the American Heart Association, American College of Cardiology, Society of Cardiovascular Angiography and Interventions, American College of Surgeons, and American Dental Association With Representation from the American College of Physicians. *Circulation*. 2007;115(6):813-818. http://circ.ahajournals.org/cgi/content/full/115/6/813. doi:10.1161/CIRCULATIONAHA.106.180944[PubMed 17224480]

Gum PA, Kottke-Marchant K, Welsh PA, et al. A Prospective, Blinded Determination of the Natural History of Aspirin Resistance Among Stable Patients With Cardiovascular Disease. *J Am Coll Cardiol*. 2003;41(6):961-965.[PubMed 12651041]

Guyatt GH, Akl EA, Crowther M, et al. Executive Summary: Antithrombotic Therapy and Prevention of Thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest*. 2012;141(2)(suppl):7-47.[PubMed 22315257]

Hahn JY, Song YB, Oh JH, et al; SMART-CHOICE Investigators. Effect of P2Y12 inhibitor monotherapy vs dual antiplatelet therapy on cardiovascular events in patients undergoing percutaneous coronary intervention: the SMART-CHOICE randomized clinical trial. *JAMA*. 2019;321(24):2428-2437. doi:10.1001/jama.2019.8146[PubMed 31237645]

Hamulyák EN, Scheres LJ, Marijnen MC, Goddijn M, Middeldorp S. Aspirin or heparin or both for improving pregnancy outcomes in women with persistent antiphospholipid antibodies and recurrent pregnancy loss. *Cochrane Database Syst Rev*. 2020;5(5):CD012852. doi:10.1002/14651858.CD012852.pub2[PubMed 32358837]

Hamulyák EN, Scheres LJJ, Goddijn M, Middeldorp S. Antithrombotic therapy to prevent recurrent pregnancy loss in antiphospholipid syndrome-What is the evidence? *J Thromb Haemost*. 2021;19(5):1174-1185. doi:10.1111/jth.15290[PubMed 33687789]

Harter HR, Burch JW, Majerus PW, et al. Prevention of thrombosis in patients on hemodialysis by low-dose aspirin. *New Eng J Med*. 1979;301(11):577-579.[PubMed 112475]

Henderson LA, Canna SW, Friedman KG, et al. American College of Rheumatology (ACR) clinical guidance for multisystem inflammatory syndrome in children associated with SARS-CoV-2 and hyperinflammation in pediatric COVID-19: Version 3. *Arthritis Rheumatol*. Published online February 3, 2022. doi:10.1002/art.42062[PubMed 35118829]

Hennekens CH. Aspirin for the secondary prevention of atherosclerotic cardiovascular disease. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed July 21, 2021.

Henrich WL, Agodoa LE, Barrett B, et al. Analgesics and the kidney: summary and recommendations to the Scientific Advisory Board of the National Kidney Foundation from an Ad Hoc Committee of the National Kidney Foundation. *Am J Kidney Dis*. 1996;27(1):162-165.[PubMed 8546133]

Hinze CH, Holzinger D, Lainka E, et al. Practice and consensus-based strategies in diagnosing and managing systemic juvenile idiopathic arthritis in Germany. *Pediatr Rheumatol Online J*. 2018;16(1):7. doi:10.1186/s12969-018-0224-2[PubMed 29357887]

Hoyte FC, Weber RW, Katial RK. Pancreatitis as a novel complication of aspirin therapy in patients with aspirin-exacerbated respiratory disease. *J Allergy Clin Immunol*. 2012;129(6):1684-1686. doi:10.1016/j.jaci.2011.12.003[PubMed 22236727]

Hsu PI, Tsai TJ. Epidemiology of upper gastrointestinal damage associated with low-dose aspirin. *Curr Pharm Des*. 2015;21(35):5049-5055. doi:10.2174/1381612821666150915104800[PubMed 26369688]

Huang WY, Saver JL, Wu YL, Lin CJ, Lee M, Ovbiagele B. Frequency of intracranial hemorrhage with low-dose aspirin in individuals without symptomatic cardiovascular disease: a systematic review and meta-analysis. *JAMA Neurol*. 2019;76(8):906-914. doi:10.1001/jamaneurol.2019.1120[PubMed 31081871]

Imazio M. Acute pericarditis: treatment and prognosis. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed September 27, 2022.

Imazio M, Demichelis B, Parrini I, et al. Day-hospital treatment of acute pericarditis. *J Am Coll Cardiol.* 2004;43(6):1042-1046. [PubMed 15028364]

International Stroke Trial Collaborative Group. The International Stroke Trial (IST): a randomised trial of aspirin, subcutaneous heparin, both, or neither among 19435 patients with acute ischaemic stroke. *Lancet*. 1997;349(9065):1569-1581.[PubMed 9174558]

Isaksson M, Jansson L. Contact allergy to Tween 80 in an inhalation suspension. *Contact Dermatitis*. 2002;47(5):312-313.[PubMed 12534540]

Ito S. Drug therapy for breast-feeding women. *N Engl J Med*. 2000;343(2):118-126.[PubMed 10891521]

Ito S, Blajchman A, Stephenson M, Eliopoulos C, Koren G. Prospective follow-up of adverse reactions in breast-fed infants exposed to maternal medication. *Am J Obstet Gynecol*. 1993;168(5):1393-1399.[PubMed 8498418]

Jamali F, Keshavarz E. Salicylate excretion in breast milk. *Int J Pharm*. 1981;8:285-290.

Jauch EC, Saver JL, Adams HP Jr, et al. Guidelines for the Early Management of Patients With Acute Ischemic Stroke: A Guideline for Healthcare Professionals From the American Heart Association/American Stroke Association. *Stroke*. 2013;44(3):870-947.[PubMed 23370205]

Jim J, Fairman MD. Percutaneous carotid artery stenting. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed May 14, 2024a.

Jim J. Transcarotid artery revascularization. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed May 14, 2024b.

Johnston SC, Amarenco P, Denison H, et al; THALES Investigators. Ticagrelor and aspirin or aspirin alone in acute ischemic stroke or TIA. *N Engl J Med*. 2020;383(3):207-217. doi:10.1056/NEJMoa1916870[PubMed 32668111]

Juurlink DN, Gosselin S, Kielstein JT, et al; EXTRIP Workgroup. Extracorporeal treatment for salicylate poisoning: systematic review and recommendations from the EXTRIP Workgroup. *Ann Emerg Med*. 2015;66(2):165-181.[PubMed 25986310]

Kalich BA, Allender JE, Hollis IB. Medication management of patients undergoing transcatheter aortic valve replacement. *Pharmacotherapy*. 2018;38(1):122-138. doi:10.1002/phar.2056[PubMed 29121410]

Källén B, Reis M. Ongoing pharmacological management of chronic pain in pregnancy. *Drugs*. 2016;76(9):915-924.[PubMed 27154242]

Kang X, Hong D, Anvari M, Tiboni M, Amin N, Gmora S. Is daily low-dose aspirin safe to take following laparoscopic Roux-en-Y gastric bypass for obesity surgery? *Obes Surg*. 2017;27(5):1261-1265. doi:10.1007/s11695-016-2462-z[PubMed 27909874]

Kannam JP, Aroesty JM, Gersh BJ. Stable ischemic heart disease: overview of care. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed August 20, 2019.

Kawakami T, Fujita A, Takeuchi S, Muto S, Soma Y. Drug-induced hypersensitivity syndrome: drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome induced by aspirin treatment of Kawasaki disease. *J Am Acad Dermatol*. 2009;60(1):146-149. doi:10.1016/j.jaad.2008.07.044[PubMed 19103366]

Kawamura N, Ito Y, Sasaki M, et al. Low-dose aspirin-associated upper gastric and duodenal ulcers in Japanese patients with no previous history of peptic ulcers. *BMC Res Notes*. 2013;6:455. doi:10.1186/1756-0500-6-455[PubMed 24215903]

K/DOQI Workgroup. K/DOQI clinical practice guidelines for cardiovascular disease in dialysis patients. *Am J Kidney Dis*. 2005;45(4)(suppl 3):S1-S153.[PubMed 15806502]

KDOQI. KDOQI Clinical Practice Guidelines and Clinical Practice Recommendations for Diabetes and Chronic Kidney Disease. *Am J Kidney Dis*. 2007;49(2)(suppl 2):S12-S154.[PubMed 17276798]

Kearon C, Akl EA, Ornelas J, et al. Antithrombotic therapy for VTE disease: CHEST guideline and expert panel report [published correction appears in *Chest*. 2016;150(4):988]. *Chest*. 2016;149(2):315-352. doi:10.1016/j.chest.2015.11.026[PubMed 26867832]

Key NS, Khorana AA, Kuderer NM, et al. A. Venous thromboembolism prophylaxis and treatment in patients with cancer: ASCO clinical practice guideline update. *J Clin Oncol.* 2020;38(5):496-520. doi:10.1200/JCO.19.01461[PubMed 31381464]

Kim YJ, Lim KH, Kim MY, et al. Cross-reactivity to acetaminophen and celecoxib according to the type of nonsteroidal anti-inflammatory drug hypersensitivity. *Allergy Asthma Immunol Res*. 2014;6(2):156-162. doi:10.4168/aair.2014.6.2.156[PubMed 24587953]

Kim BK, Hong SJ, Cho YH, et al; TICO Investigators. Effect of ticagrelor monotherapy vs ticagrelor with aspirin on major bleeding and cardiovascular events in patients with acute coronary syndrome: the TICO randomized clinical trial. *JAMA*. 2020;323(23):2407-2416. doi:10.1001/jama.2020.7580[PubMed 32543684]

Kirthi V, Derry S, Moore RA. Aspirin with or without an antiemetic for acute migraine headaches in adults. *Cochrane Database Syst Rev*. 2013;2013(4):CD008041. doi:10.1002/14651858.CD008041.pub3[PubMed 23633350]

Klar H, Sotošek N, Šelb J, Košnik M. Selective hypersensitivity to a single nonsteroidal anti-inflammatory drug. *Acta Dermatovenerol Alp Pannonica Adriat*. 2019;28(3):97-101.[PubMed 31545385]

Kleindorfer DO, Towfighi A, Chaturvedi S, et al. 2021 Guideline for the prevention of stroke in patients with stroke and transient ischemic attack: A guideline from the American Heart Association/American Stroke Association. *Stroke*. 2021;52(7):e364-e467. doi:10.1161/STR.0000000000000375[PubMed 34024117]

Kliegman RM, St. Geme J, eds. *Nelson Textbook of Pediatrics*. 21st ed. Saunders Elsevier; 2020.

Knowles SR, Drucker AM, Weber EA, Shear NH. Management options for patients with aspirin and nonsteroidal antiinflammatory drug sensitivity. *Ann Pharmacother*. 2007;41(7):1191-1200. doi:10.1345/aph.1K023[PubMed 17609236]

Koren G. Hearing loss in a woman on aspirin: the silent pharmacokinetic parameter. *Ther Drug Monit*. 2009;31(1):1-2. doi:10.1097/FTD.0b013e3181951ed9[PubMed 19155962]

Kowalski ML, Agache I, Bavbek S, et al. Diagnosis and management of NSAID-exacerbated respiratory disease (N-ERD)-a EAACI position paper. *Allergy*. 2019;74(1):28-39. doi:10.1111/all.13599[PubMed 30216468]

Kowalski ML, Asero R, Bavbek S, et al. Classification and practical approach to the diagnosis and management of hypersensitivity to nonsteroidal anti-inflammatory drugs. *Allergy*. 2013;68(10):1219-1232. doi:10.1111/all.12260[PubMed 24117484]

Kowalski ML, Makowska JS, Blanca M, et al. Hypersensitivity to nonsteroidal anti-inflammatory drugs (NSAIDs) - classification, diagnosis and management: review of the EAACI/ENDA(#) and GA2LEN/HANNA*. *Allergy*. 2011;66(7):818-829. doi:10.1111/j.1398-9995.2011.02557.x[PubMed 21631520]

Kowalski ML, Makowska JS. Seven steps to the diagnosis of NSAIDs hypersensitivity: how to apply a new classification in real practice? *Allergy Asthma Immunol Res*. 2015;7(4):312-320. doi:10.4168/aair.2015.7.4.312[PubMed 25749768]

Kulik A, Ruel M, Jneid H, et al; American Heart Association Council on Cardiovascular Surgery and Anesthesia. Secondary prevention after coronary artery bypass graft surgery: a scientific statement from the American Heart Association. *Circulation*. 2015;131(10):927-964. doi:10.1161/CIR.0000000000000182[PubMed 25679302]

Kumbhani DJ, Cannon CP, Beavers CJ, et al. 2020 ACC expert consensus decision pathway for anticoagulant and antiplatelet therapy in patients with atrial fibrillation or venous thromboembolism undergoing percutaneous coronary intervention or with atherosclerotic cardiovascular disease: a report of the American College of Cardiology Solution Set Oversight Committee. *J Am Coll Cardiol*. 2021;77(5):629-658. doi:10.1016/j.jacc.2020.09.011[PubMed 33250267]

Kuno T, Takagi H, Sugiyama T, et al. Antithrombotic strategies after transcatheter aortic valve implantation: insights from a network meta-analysis [published online October 14, 2019]. *Catheter Cardiovasc Interv*. doi:10.1002/ccd.28498[PubMed 31609071]

Kwolek CJ, Jaff MR, Leal JI, et al. Results of the ROADSTER multicenter trial of transcarotid stenting with dynamic flow reversal. *J Vasc Surg*. 2015;62(5):1227-1234. doi:10.1016/j.jvs.2015.04.460[PubMed 26506270]

Lafrance JP, Miller DR. Selective and non-selective non-steroidal anti-inflammatory drugs and the risk of acute kidney injury. *Pharmacoepidemiol DrugSaf*. 2009;18(10):923-931. doi:10.1002/pds.1798[PubMed 19585463]

Laidlaw TM, Cahill KN. Current knowledge and management of hypersensitivity to aspirin and NSAIDs. *J Allergy Clin Immunol Pract*. 2017;5(3):537-545. doi:10.1016/j.jaip.2016.10.021[PubMed 28483309]

Landolfi R, Marchioli R, Kutti J, et al; European Collaboration on Low-Dose Aspirin in Polycythemia Vera Investigators. Efficacy and safety of low-dose aspirin in polycythemia vera. *N Engl J Med*. 2004;350(2):114-124. doi:10.1056/NEJMoa035572[PubMed 14711910]

Lansberg MG, O'Donnell MJ, Khatri P, et al. Antithrombotic and thrombolytic therapy for ischemic stroke: antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians evidence-based clinical practice guidelines. *Chest*. 2012;141(2)(suppl):e601S-e636S. doi:10.1378/chest.11-2302[PubMed 22315273]

Larocca A, Cavallo F, Bringhen S, et al. Aspirin or enoxaparin thromboprophylaxis for patients with newly diagnosed multiple myeloma treated with lenalidomide. *Blood*. 2012;119(4):933-939; quiz 1093. doi:10.1182/blood-2011-03-344333[PubMed 21835953]

Lawton JS, Tamis-Holland JE, Bangalore S, et al. 2021 ACC/AHA/SCAI guideline for coronary artery revascularization: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2022;79(2):e21-e129. doi:10.1016/j.jacc.2021.09.006[PubMed 34895950]

LeBlanc JG, Sett SS, Vince DJ. Antiplatelet Therapy in Children With Left-Sided Mechanical Prostheses. *Eur J Cardiothorac Surg*. 1993;7(4):211-215.[PubMed 8481259]

Leon MB, Smith CR, Mack M, et al; PARTNER Trial Investigators. Transcatheter aortic-valve implantation for aortic stenosis in patients who cannot undergo surgery. *N Engl J Med*. 2010;363(17):1597-1607. doi:10.1056/NEJMoa1008232[PubMed 20961243]

Leon MB, Smith CR, Mack MJ, et al; PARTNER 2 Investigators. Transcatheter or surgical aortic-valve replacement in intermediate-risk patients. *N Engl J Med*. 2016;374(17):1609-1620. doi:10.1056/NEJMoa1514616[PubMed 27040324]

Leventhal LJ, Kuritsky L, Ginsburg R, Bomalaski JS. Salicylate-induced rhabdomyolysis. *Am J Emerg Med*. 1989;7(4):409-410. doi:10.1016/0735-6757(89)90049-1[PubMed 2735989]

Levy G. Clinical Pharmacokinetics of Aspirin. *Pediatrics*. 1978;62(5, pt 2)(suppl):867-872.[PubMed 724339]

Levy G, Procknal JA, Garrettson LK. Distribution of salicylate between neonatal and maternal serum at diffusion equilibrium. *Clin Pharmacol Ther*. 1975;18(2):210-214.[PubMed 1157445]

LeWinter MM. Pericardial complications of myocardial infarction. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed August 2, 2022.

Li L, Li W, Chen CZ, Yi ZH, Zhou YY. Is aspirin use associated with age-related macular degeneration? A meta-analysis. *J Clin Pharm Ther*. 2015;40(2):144-154.[PubMed 25475899]

Lipton RB, Goldstein J, Baggish JS, Yataco AR, Sorrentino JV, Quiring JN. Aspirin is efficacious for the treatment of acute migraine. *Headache*. 2005;45(4):283-292. doi:10.1111/j.1526-4610.2005.05065.x[PubMed 15836564]

Livio M, Benigni A, Viganò G, Mecca G, Remuzzi G. Moderate doses of aspirin and risk of bleeding in renal failure. *Lancet*. 1986;1(8478):414-416. doi:10.1016/s0140-6736(86)92372-x[PubMed 2868341]

Lucente P, Iorizzo M, Pazzaglia M. Contact sensitivity to Tween 80 in a child. *Contact Dermatitis*. 2000;43(3):172.[PubMed 10985636]

Lyman GH, Carrier M, Ay C, et al. American Society of Hematology 2021 guidelines for management of venous thromboembolism: prevention and treatment in patients with cancer. *Blood Adv.* 2021;5(4):927-974. doi:10.1182/bloodadvances.2020003442[PubMed 33570602]

Maalouf R, Mosley M, James Kallail K, Kramer KM, Kumar G. A comparison of salicylic acid levels in normal subjects after rectal versus oral dosing. *Acad Emerg Med*. 2009;16(2):157-161. doi:10.1111/j.1553-2712.2008.00318.x[PubMed 19076101]

MacDonald TM, Wei L. Effect of ibuprofen on cardioprotective effect of aspirin. *Lancet*. 2003;361(9357):573-574. doi:10.1016/s0140-6736(03)12509-3[PubMed 12598144]

MacGregor EA, Dowson A, Davies PT. Mouth-dispersible aspirin in the treatment of migraine: a placebo-controlled study. *Headache*. 2002;42(4):249-255. doi:10.1046/j.1526-4610.2002.02076.x[PubMed 12010380]

Magee LA, Smith GN, Bloch C, et al. Guideline no. 426: hypertensive disorders of pregnancy: diagnosis, prediction, prevention, and management. *J Obstet Gynaecol Can*. 2022;44(5):547-571.e1. doi:10.1016/j.jogc.2022.03.002[PubMed 35577426]

Marcaccio CL, Patel PB, Liang P, et al. Efficacy and safety of perioperative dual antiplatelet therapy with ticagrelor versus clopidogrel in carotid artery stenting. *J Vasc Surg*. 2022;75(4):1293-1303.e8. doi: 10.1016/j.jvs.2021.09.045[PubMed 34655685]

Martin E, Vickers B, Landau R, Reece-Stremtan S. ABM clinical protocol #28, peripartum analgesia and anesthesia for the breastfeeding mother. *Breastfeed Med*. 2018;13(3):164-171.[PubMed 29595994]

Mauri L, Kereiakes DJ, Yeh RW, et al; DAPT Study Investigators. Twelve or 30 months of dual antiplatelet therapy after drug-eluting stents. *N Engl J Med*. 2014;371(23):2155-2166. doi:10.1056/NEJMoa1409312[PubMed 25399658]

McCrindle BW, Rowley AH, Newburger JW, et al. Diagnosis, treatment, and long-term management of Kawasaki Disease: a scientific statement for health professionals from the American Heart Association. *Circulation*. 2017;135(17):e927-e999.[PubMed 28356445]

McGovern MC, Glasgow JF, Stewart MC. Lesson of the week: Reye's syndrome and aspirin: lest we forget. *BMJ*. 2001;322(7302):1591-1592. doi:10.1136/bmj.322.7302.1591[PubMed 11431304]

McMullin MF, Harrison CN, Ali S, et al; BSH Committee. A guideline for the diagnosis and management of polycythaemia vera. A British Society for Haematology guideline. *Br J Haematol*. 2019;184(2):176-191. doi:10.1111/bjh.15648[PubMed 30478826]

McNeil JJ, Wolfe R, Woods RL, et al. Effect of aspirin on cardiovascular events and bleeding in the healthy elderly. *N Engl J Med*. 2018;379(16):1509-1518. doi:10.1056/NEJMoa1805819[PubMed 30221597]

McQuilten ZK, Thao LTP, Pasricha SR, et al. Effect of low-dose aspirin versus placebo on incidence of anemia in the elderly: a secondary analysis of the aspirin in reducing events in the elderly trial. *Ann Intern Med*. 2023;176(7):913-921. doi:10.7326/M23-0675[PubMed 37335992]

Medeiros FB, de Andrade AC, Angelis GA, et al. Bleeding Evaluation During Single Tooth Extraction in Patients With Coronary Artery Disease and Acetylsalicylic Acid Therapy Suspension: A Prospective, Double-Blinded, and Randomized Study. *J Oral Maxillofac Surg*. 2011;69(12):2949-2955.[PubMed 21802823]

Mehran R, Baber U, Sharma SK, et al. Ticagrelor with or without aspirin in high-risk patients after PCI. *N Engl J Med*. 2019;381(21):2032-2042. doi:10.1056/NEJMoa1908419[PubMed 31556978]

Mehta SR, Yusuf S, Peters RJ, et al; Clopidogrel in Unstable Angina to Prevent Recurrent Events Trial (CURE) Investigators. Effects of pretreatment with clopidogrel and aspirin followed by long-term therapy in patients undergoing percutaneous coronary intervention: the PCI-CURE study. *Lancet*. 2001;358(9281):527-533.[PubMed 11520521]

Meschia JF, Bushnell C, Boden-Albala B, et al; American Heart Association Stroke Council; Council on Cardiovascular and Stroke Nursing; Council on Clinical Cardiology; Council on Functional Genomics and Translational Biology; Council on Hypertension. Guidelines for the primary prevention of stroke: a statement for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke*. 2014;45(12):3754-832. doi:10.1161/STR.0000000000000046[PubMed 25355838]

Meyers RS, Thackray J, Matson KL, et al. *K*ey Potentially *I*nappropriate *D*rugs in Pediatrics: The KIDs List. *J Pediatr Pharmacol Ther*. 2020;25(3):175-191.[PubMed 32265601]

Mitrov-Winkelmolen L, van Buul-Gast MW, Swank DJ, Overdiek HWPM, van Schaik RHN, Touw DJ. The effect of Roux-en-Y gastric bypass surgery in morbidly obese patients on pharmacokinetics of (acetyl)salicylic acid and omeprazole: the ERY-PAO study. *Obes Surg*. 2016;26(9):2051-2058. doi:10.1007/s11695-016-2065-8[PubMed 26797705]

Monagle P, Chan A, Goldenberg NA, et al. Antithrombotic Therapy in Neonates and Children: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines (9th Edition). *Chest*. 2012;141(2)(suppl):e737-e801.[PubMed 22315277]

Monagle P, Cochrane A, Roberts R, et al. A Multicenter, Randomized Trial Comparing Heparin/Warfarin and Acetylsalicylic Acid as Primary Thromboprophylaxis for 2 Years after the Fontan Procedure in Children. *J Am Coll Cardiol*. 2011;58(6):645-651.[PubMed 21798429]

Naimi AI, Perkins NJ, Sjaarda LA, et al. The effect of preconception-initiated low-dose aspirin on human chorionic gonadotropin-detected pregnancy, pregnancy loss, and live birth: per protocol analysis of a randomized trial. *Ann Intern Med*. 2021;174(5):595-601. doi:10.7326/M20-0469[PubMed 33493011]

National Institute for Health and Care Excellence. Drug allergy: diagnosis and management. https://www.nice.org.uk/guidance/cg183 Published September 3, 2014. Accessed November 3, 2021.

National Institutes of Health (NIH). COVID-19 Treatment Guidelines Panel. Therapeutic management of hospitalized pediatric patients with multisystem inflammatory syndrome in children (MIS-C) (with discussion on multisystem inflammatory syndrome in adults [MIS-A]). Updated February 24, 2022. Accessed March 7, 2022.

Newburger JW, Takahashi M, Gerber MA, et al. Diagnosis, treatment, and long-term management of Kawasaki disease: a statement for health professionals from the Committee on Rheumatic Fever, Endocarditis, and Kawasaki Disease, Council on Cardiovascular Disease in the Young, American Heart Association. *Pediatrics*. 2004;114(6):1708-1733.[PubMed 15574639]

Nkomo VT, Konkle BA. Antithrombotic therapy for surgical bioprosthetic valves and surgical valve repair. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed June 22, 2022.

Norgard NB, Monte SV, Fernandez SF, Ma Q. Aspirin responsiveness changes in obese patients following bariatric surgery. *Cardiovasc Ther*. 2017;35(4). doi:10.1111/1755-5922.12268[PubMed 28444860]

O'Donoghue ML, Murphy SA, Sabatine MS. The safety and efficacy of aspirin discontinuation on a background of a $P2Y_{12}$ inhibitor in patients after percutaneous coronary intervention: a systematic review and meta-analysis. *Circulation*. 2020;142(6):538-545. doi:10.1161/CIRCULATIONAHA.120.046251[PubMed 32551860]

O'Gara PT, Kushner FG, Ascheim DD, et al; American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. 2013 ACCF/AHA guideline for the management of ST-elevation myocardial infarction: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2013;127(4):e362-e425. doi:10.1161/CIR.0b013e3182742cf6[PubMed 23247304]

O'Mahony D, Cherubini A, Guiteras AR, et al. STOPP/START criteria for potentially inappropriate prescribing in older people: version 3. *Eur Geriatr Med*. 2023;14(4):625-632. doi:10.1007/s41999-023-00777-y[PubMed 37256475]

Oliveira-Filho J, Mullen MT. Early antithrombotic treatment of acute ischemic stroke and transient ischemic attack. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed November 10, 2022.

Østensen M. Nonsteroidal Anti-inflammatory Drugs During Pregnancy. *Scand J Rheumatol Suppl*. 1998;107:128-132.[PubMed 9759151]

Otto CM, Kumbhani DJ, Alexander KP, et al. 2017 ACC expert consensus decision pathway for transcatheter aortic valve replacement in the management of adults with aortic stenosis: a report of the American College of Cardiology Task Force on Clinical Expert Consensus Documents. *J Am Coll Cardiol*. 2017;69(10):1313-1346. doi:10.1016/j.jacc.2016.12.006[PubMed 28063810]

Otto CM, Nishimura RA, Bonow RO, et al. 2020 ACC/AHA guideline for the management of patients with valvular heart disease: a report of the American College of Cardiology/American Heart Association Joint Committee on clinical practice guidelines. *Circulation*. 2021;143(5):e72-e227. doi:10.1161/CIR.0000000000000923[PubMed 33332150]

Paikin JS, Wright DS, Crowther MA, Mehta SR, Eikelboom JW. Triple antithrombotic therapy in patients with atrial fibrillation and coronary artery stents. *Circulation*. 2010;121(18):2067-2070. doi:10.1161/CIRCULATIONAHA.109.924944[PubMed 20458022]

Palmisano PA, Cassady G. Salicylate exposure in the perinate. *JAMA*. 1969;209(4):556-558.[PubMed 5819427]

Palumbo A, Cavo M, Bringhen S, et al. Aspirin, warfarin, or enoxaparin thromboprophylaxis in patients with multiple myeloma treated with thalidomide: a phase III, open-label, randomized trial. *J Clin Oncol.* 2011;29(8):986-993. doi:10.1200/JCO.2010.31.6844[PubMed 21282540]

Pascale S, Petrucci G, Dragani A, et al. Aspirin-insensitive thromboxane biosynthesis in essential thrombocythemia is explained by accelerated renewal of the drug target. *Blood*. 2012;119(15):3595-3603. doi:10.1182/blood-2011-06-359224[PubMed 22234683]

Poh J, Knowles S. Safety of 5-aminosalicylic acid derivatives in patients with sensitivity to acetylsalicylic acid and nonsteroidal anti-inflammatory drugs. *Can J Hosp Pharm*. 2014;67(1):35-38. doi:10.4212/cjhp.v67i1.1318[PubMed 24634525]

Poon LC, Shennan A, Hyett JA, et al. The International Federation of Gynecology and Obstetrics (FIGO) initiative on pre-eclampsia: a pragmatic guide for first-trimester screening and prevention. *Int J Gynaecol Obstet*. 2019;145(suppl 1):1-33. doi:10.1002/ijgo.12802[PubMed 31111484]

Powers WJ, Rabinstein AA, Ackerson T, et al. Guidelines for the early management of patients with acute ischemic stroke: 2019 update to the 2018 guidelines for the early management of acute ischemic stroke: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke*. 2019;50(12):e344-e418. doi:10.1161/STR.0000000000000211[PubMed 31662037]

Puskas J, Gerdisch M, Nichols D, et al; PROACT Investigators. Reduced anticoagulation after mechanical aortic valve replacement: interim results from the prospective randomized On-X valve anticoagulation clinical trial randomized Food and Drug Administration investigational device exemption trial. *J Thorac Cardiovasc Surg.* 2014;147(4):1202-1210; discussion 1210-1211. doi:10.1016/j.jtcvs.2014.01.004[PubMed 24512654]

Rajkumar SV, Hayman SR, Lacy MQ, et al. Combination therapy with lenalidomide plus dexamethasone (Rev/Dex) for newly diagnosed myeloma. *Blood*. 2005;106(13):4050-4053. doi:10.1182/blood-2005-07-2817[PubMed 16118317]

Reardon MJ, Van Mieghem NM, Popma JJ, et al; SURTAVI Investigators. Surgical or transcatheter aortic-valve replacement in intermediate-risk patients. *N Engl J Med*. 2017;376(14):1321-1331. doi:10.1056/NEJMoa1700456[PubMed 28304219]

Reece-Stremtan S, Campos M, Kokajko L; Academy of Breastfeeding Medicine. ABM clinical protocol #15: analgesia and anesthesia for the breastfeeding mother, revised 2017. *Breastfeed Med*. 2017;12(9):500-506.[PubMed 29624435]

Refer to manufacturer's labeling.

Regitz-Zagrosek V, Roos-Hesselink JW, Bauersachs J, et al. 2018 ESC Guidelines for the management of cardiovascular diseases during pregnancy. *Eur Heart J*. 2018;39(34):3165-3241.[PubMed 30165544]

RHD Australia. The 2020 Australian guideline for prevention, diagnosis and management of acute rheumatic fever and rheumatic heart disease. 3rd edition. https://www.rhdaustralia.org.au/system/files/fileuploads/arf_rhd_guidelines_3rd_edition_final.pdf

Ringold S, Angeles-Han ST, Beukelman T, et al. 2019 American College of Rheumatology/Arthritis Foundation guideline for the treatment of juvenile idiopathic arthritis: therapeutic approaches for non-systemic polyarthritis, sacroiliitis, and enthesitis. *Arthritis Care Res (Hoboken)*. 2019;71(6):717-734. doi:10.1002/acr.23870[PubMed 31021516]

Ringold S, Weiss PF, Colbert RA, et al. Childhood Arthritis and Rheumatology Research Alliance consensus treatment plans for new-onset polyarticular juvenile idiopathic arthritis. *Arthritis Care Res (Hoboken)*. 2014;66(7):1063-1072. doi:10.1002/acr.22259[PubMed 24339215]

Rolnik DL, Wright D, Poon LC, et al. Aspirin versus placebo in pregnancies at high risk for preterm preeclampsia. *N Engl J Med.* 2017;377(7):613-622. doi:10.1056/NEJMoa1704559[PubMed 28657417]

Rose CD, Doughty RA. Pharmacological management of juvenile rheumatoid arthritis. *Drugs*. 1992;43(6):849-863. doi:10.2165/00003495-199243060-00005[PubMed 1379157]

Rosenberg J, Benowitz NL, Pond S. Pharmacokinetics of drug overdose. *Clin Pharmacokinet*. 1981;6(3):161-192.[PubMed 7016383]

Rothwell PM, Wilson M, Elwin CE, et al. Long-term effect of aspirin on colorectal cancer incidence and mortality: 20-year follow-up of five randomized trials. *Lancet*. 2010;376(9754):1741-1750. doi:10.1016/S0140-6736(10)61543-7[PubMed 20970847]

Rymark P, Berntorp E, Nordsjö P, Liedholm H, Melander A, Gennser G. Low-dose aspirin to pregnant women: single dose pharmacokinetics and influence of short term treatment on bleeding time. *J Perinat Med*. 1994;22(3):205-211.[PubMed 7823260]

Sachs HC; Committee On Drugs. The transfer of drugs and therapeutics into human breast milk: an update on selected topics. *Pediatrics*. 2013;132(3):e796-e809.[PubMed 23979084]

Sai Y, Kusaka A, Imanishi K, et al. A randomized, quadruple crossover single-blind study on immediate action of chewed and unchewed low-dose acetylsalicylic acid tablets in healthy volunteers. *J Pharm Sci*. 2011;100(9):3884-3891. doi:10.1002/jps.22602[PubMed 21544823]

Saldanha IJ, Cao W, Bhuma MR, et al. Management of primary headaches during pregnancy, postpartum, and breastfeeding: a systematic review. *Headache*. 2021;61(1):11-43. doi:10.1111/head.14041[PubMed 33433020]

Sammaritano LR, Bermas BL, Chakravarty EE, et al. 2020 American College of Rheumatology guideline for the management of reproductive health in rheumatic and musculoskeletal diseases. *Arthritis Rheumatol*. 2020;72(4):529-556. doi:10.1002/art.41191[PubMed 32090480]

Sandercock PA, Counsell C, Tseng MC, Cecconi E. Oral antiplatelet therapy for acute ischaemic stroke. *Cochrane Database Syst Rev*. 2014;(3):CD000029. doi:10.1002/14651858.CD000029.pub3[PubMed 24668137]

Savage R. Cyclo-oxygenase-2 inhibitors: when should they be used in the elderly? *Drugs Aging*. 2005;22(3):185-200. doi:10.2165/00002512-200522030-00001.[PubMed 15813652]

Schwedt TJ, Garza Ivan. Acute treatment of migraine in adults. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed August 9, 2021.

Segal R, Lubart E, Leibovitz A, Iaina A, Caspi D. Renal effects of low dose aspirin in elderly patients. *Isr Med Assoc J*. 2006;8(10):679-682.[PubMed 17125112]

Serebruany VL, Steinhubl SR, Berger PB, et al. Analysis of risk of bleeding complications after different doses of aspirin in 192,036 patients enrolled in 31 randomized controlled trials. *Am J Cardiol*. 2005;95(10):1218-1222. doi:10.1016/j.amjcard.2005.01.049[PubMed 15877994]

Serra AJ, McNicholas KW, Olivier HF Jr, et al. The Choice of Anticoagulation in Pediatric Patients With the St. Jude Medical Valve Prostheses. *J Cardiovasc Surg*. 1987;28(5):588-591.[PubMed 3654744]

Shah S, Banh ET, Koury K, Bhatia G, Nandi R, Gulur P. Pain management in pregnancy: multimodal approaches. *Pain Res Treat*. 2015;2015:987483.[PubMed 26448875]

Shanmugalingam R, Wang X, Münch G, et al. A pharmacokinetic assessment of optimal dosing, preparation, and chronotherapy of aspirin in pregnancy. *Am J Obstet Gynecol*. 2019;221(3):255.e1-e255.[PubMed 31051121]

Shelley WB, Talanin N, Shelley ED. Polysorbate 80 hypersensitivity. *Lancet*. 1995;345(8980):1312-1313.[PubMed 7746084]

Silver R, Craigo S, Porter F, Osmundson S, Norton ME; Society for Maternal-Fetal Medicine (SMFM). Society for Maternal-Fetal Medicine (SMFM) consult series #64: systemic lupus erythematosus in pregnancy. *Am J Obstet Gynecol.* 2022:S0002-9378(22)00722-0. doi:10.1016/j.ajog.2022.09.001[PubMed 36084704]

Simes J, Becattini C, Agnelli G, et al; INSPIRE Study Investigators (International Collaboration of Aspirin Trials for Recurrent Venous Thromboembolism). Aspirin for the prevention of recurrent venous thromboembolism: the INSPIRE collaboration. *Circulation*. 2014;130(13):1062-1071. doi:10.1161/CIRCULATIONAHA.114.008828[PubMed 25156992]

Smith CR, Leon MB, Mack MJ, et al; PARTNER Trial Investigators. Transcatheter versus surgical aortic-valve replacement in high-risk patients. *N Engl J Med*. 2011;364(23):2187-2198. doi:10.1056/NEJMoa1103510[PubMed 21639811]

Solymar L, Rao PS, Mardini MK, et al. Prosthetic Valves in Children and Adolescents. *Am Heart J*. 1991;121(2, pt 1):557-568.[PubMed 1990763]

Spigset O, Hagg S. Analgesics and Breast-Feeding: Safety Considerations. *Paediatr Drugs*. 2000;2(3):223-238.[PubMed 10937472]

Steer A, Gibofsky A. Acute rheumatic fever: treatment and prevention. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed October 22, 2019.

Stevens SM, Woller SC, Kreuziger LB, et al. Antithrombotic therapy for VTE disease: second update of the CHEST guideline and expert panel report. *Chest*. 2021;160(6):e545-e608. doi:10.1016/j.chest.2021.07.055[PubMed 34352278]

Stevenson DD, Simon RA, Lumry WR, Mathison DA. Adverse reactions to tartrazine. *J Allergy Clin Immunol*. 1986;78(1 Pt 2):182-191. doi:10.1016/0091-6749(86)90011-4[PubMed 3722646]

Stone GW, Lindenfeld J, Abraham WT, et al. Transcatheter mitral-valve repair in patients with heart failure. *N Engl J Med*. 2018;379(24):2307-2318. doi:10.1056/NEJMoa1806640[PubMed 30280640]

Summaria F, Giannico MB, Talarico GP, Patrizi R. Antiplatelet Therapy in Hemodialysis Patients Undergoing Percutaneous Coronary Interventions. *Nephrourol Mon*. 2015;7(4):e28099.[PubMed 26528445]

Sverdén E, Mattsson F, Sondén A, et al. Risk factors for marginal ulcer after gastric bypass surgery for obesity: a population-based cohort study. *Ann Surg*. 2016;263(4):733-737. doi:10.1097/SLA.0000000000001300[PubMed 26106845]

Tefferi A. Polycythemia vera and secondary polycythemia: treatment and prognosis. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed September 9, 2024.

Tefferi A, Barbui T. Polycythemia vera and essential thrombocythemia: 2017 update on diagnosis, risk-stratification, and management. *Am J Hematol*. 2017;92(1):94-108. doi:10.1002/ajh.24607[PubMed 27991718]

Terlemez S, Demir F, Bulut Y, et al. DRESS syndrome developed related to acetylsalicylic acid use. *Pediatr Allergy Immunol*. 2016;27(2):227-230. doi:10.1111/pai.12484[PubMed 26928755]

Terragna A, Spirito L. Thrombocytopenic purpura in an infant after administration of acetylsalicylic acid to the wet-nurse. *Minerva Pediatr*. 1967;19(13):613-616.[PubMed 6069440]

Terzioğlu K, Sancar Ö, Ekerbiçer HÇ, Öztürk RT, Epöztürk K. Tolerability to paracetamol and preferential COX-2 inhibitors in patients with cross-reactive nonsteroidal anti-inflammatory drugs hypersensitivity. *Asia Pac Allergy*. 2020;10(3):e29. doi:10.5415/apallergy.2020.10.e29[PubMed 32789114]

Turan TN, Gutierrez J. Intracranial large artery atherosclerosis: Treatment and prognosis. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed May 8, 2024.

Turner JM, Robertson NT, Hartel G, Kumar S. Impact of low-dose aspirin on adverse perinatal outcome: meta-analysis and meta-regression. *Ultrasound Obstet Gynecol*. 2020;55(2):157-169. doi:10.1002/uog.20859[PubMed 31479546]

Unsworth J, d'Assis-Fonseca A, Beswick DT, Blake DR. Serum salicylate levels in a breast fed infant. *Ann Rheum Dis*. 1987;46(8):638-639.[PubMed 3662653]

US Food and Drug Administration (FDA). FDA Drug Safety Communication. MedWatch. FDA recommends avoiding use of NSAIDs in pregnancy at 20 weeks or later because they can result in low amniotic fluid. https://www.fda.gov/drugs/drug-safety-and-availability/fda-recommends-avoiding-use-nsaids-pregnancy-20-weeks-or-later-because-they-can-result-low-amniotic. Published October 16, 2020. Accessed October 21, 2020.

US Preventive Services Task Force (USPSTF). *Aspirin for the Prevention of Cardiovascular Disease: Clinical Summary*. AHRQ Publication No.09-05129-EF-3, March 2009. Agency for Healthcare Research and Quality, Rockville, MD. http://www.uspreventiveservicestaskforce.org/uspstf09/aspirincvd/aspcvdsum.htm

US Preventive Services Task Force (USPSTF); Davidson KW, Barry MJ, Mangione CM, et al. Aspirin use to prevent cardiovascular disease: US Preventive Services Task Force recommendation statement. *JAMA*. 2022;327(16):1577-1584. doi:10.1001/jama.2022.4983[PubMed 35471505]

Vannucchi AM, Barbui T, Cervantes F, et al; ESMO Guidelines Committee. Philadelphia chromosome-negative chronic myeloproliferative neoplasms: ESMO clinical practice guidelines for diagnosis, treatment and follow-up. *Ann Oncol.* 2015;26(suppl 5):v85-v99. doi:10.1093/annonc/mdv203[PubMed 26242182]

Vásquez LDM, Silva DL, Ramírez LF, Olaya M, Serrano CD. Descriptive analysis of cross-reactive anaphylaxis as a different clinical subtype of nonsteroidal anti-inflammatory drug (NSAID) hypersensitivity. *Int Arch Allergy Immunol*. 2021;182(2):131-138. doi:10.1159/000510335[PubMed 32957109]

Vazalore (aspirin) [prescribing information]. Sparta, NJ: PLx Pharma Inc; April 2021.

Virani SS, Newby LK, Arnold SV, et al. 2023 AHA/ACC/ACCP/ASPC/NLA/PCNA guideline for the management of patients with chronic coronary disease: a report of the American Heart Association/American College of Cardiology Joint Committee on Clinical Practice Guidelines. *Circulation*. 2023;148(9):e9-e119. doi:10.1161/CIR.0000000000001168[PubMed 37471501]

Vranckx P, Valgimigli M, Jüni P, et al; GLOBAL LEADERS Investigators. Ticagrelor plus aspirin for 1 month, followed by ticagrelor monotherapy for 23 months vs aspirin plus clopidogrel or ticagrelor for 12 months, followed by aspirin monotherapy for 12 months after implantation of a drug-eluting stent: a multicentre, open-label, randomised superiority trial. *Lancet*. 2018;392(10151):940-949. doi:10.1016/S0140-6736(18)31858-0[PubMed 30166073]

Walker MD, Marler JR, Goldstein M, et al; Executive Committee for the Asymptomatic Carotid Atherosclerosis Study. Endarterectomy for asymptomatic carotid artery stenosis. *JAMA*. 1995;273(18):1421-1428. doi:10.1001/jama.1995.03520420037035[PubMed 7723155]

Wallentin L, Becker RC, Budaj A, et al; PLATO Investigators. Ticagrelor versus clopidogrel in patients with acute coronary syndromes. *N Engl J Med*. 2009;361(11):1045-1057. doi:10.1056/NEJMoa0904327[PubMed 19717846]

Watanabe H, Domei T, Morimoto T, et al; STOPDAPT-2 Investigators. Effect of 1-month dual antiplatelet therapy followed by clopidogrel vs 12-month dual antiplatelet therapy on cardiovascular and bleeding events in patients receiving PCI: the STOPDAPT-2 randomized clinical trial. *JAMA*. 2019;321(24):2414-2427. doi:10.1001/jama.2019.8145[PubMed 31237644]

Whelton A. Renal and related cardiovascular effects of conventional and COX-2-specific NSAIDs and non-NSAID analgesics. *Am J Ther*. 2000;7(2):63-74.[PubMed 11319575]

White AA, Stevenson DD. Aspirin-exacerbated respiratory disease. *N Engl J Med*. 2018;379(11):1060-1070. doi:10.1056/NEJMra1712125[PubMed 30207919]

Wiviott SD, Braunwald E, McCabe CH, et al; TRITON-TIMI 38 Investigators. Prasugrel versus clopidogrel in patients with acute coronary syndromes. *N Engl J Med*. 2007;357(20):2001-2015. doi:10.1056/NEJMoa0706482[PubMed 17982182]

Woessner KM. Crossreacting drugs and chemicals. *Clin Rev Allergy Immunol*. 2003;24(2):149-158. doi:10.1385/CRIAI:24:2:149[PubMed 12668895]

Wongrakpanich S, Wongrakpanich A, Melhado K, Rangaswami J. A comprehensive review of non-steroidal anti-inflammatory drug use in the elderly. *Aging Dis*. 2018;9(1):143-150. doi:10.14336/AD.2017.0306[PubMed 29392089]

World Health Organization (WHO). Breastfeeding and maternal medication, recommendations for drugs in the eleventh WHO model list of essential drugs. https://apps.who.int/iris/handle/10665/62435. Published 2002.

World Health Organization (WHO). Rheumatic fever and rheumatic heart disease: report of a WHO Expert Consultation. Geneva, Switzerland: World Health Organization; 2004. http://www.who.int/cardiovascular_diseases/resources/en/cvd_trs923.pdf. Accessed December 15, 2015.

Ye X, Fu J, Yang Y, Chen S. Dose-risk and duration-risk relationships between aspirin and colorectal cancer: a meta-analysis of published cohort studies. *PLoS One*. 2013;8(2):e57578. doi:10.1371/journal.pone.0057578[PubMed 23451245]

Yusuf S, Zhao F, Mehta SR, Chrolavicius S, Tognoni G, Fox KK; Clopidogrel in Unstable Angina to Prevent Recurrent Events Trial Investigators. Effects of clopidogrel in addition to aspirin in patients with acute coronary syndromes without ST-segment elevation [published corrections appear in *N Engl J Med*. 2001;345(20):1506; *N Engl J Med*. 2001;345(23):1716]. *N Engl J Med*. 2001;345(7):494-502. doi:10.1056/NEJMoa010746[PubMed 11519503]

Zhang X, Donnan PT, Bell S, Guthrie B. Non-steroidal anti-inflammatory drug induced acute kidney injury in the community dwelling general population and people with chronic kidney disease: systematic review and meta-analysis. *BMC Nephrol*. 2017;18(1):256. doi:10.1186/s12882-017-0673-8[PubMed 28764659]

References: International

Lag injection (aspirin) [product information]. Hsinchu County, Taiwan: Taiyu Chemical Pharmaceutical Factory Co Ltd; December 2015.

Brand Names: US

Ascriptin [OTC]; Aspirin 81 [OTC] [DSC]; Aspirin Adult Low Dose [OTC]; Aspirin Adult Low Strength [OTC]; Aspirin Adult [OTC] [DSC]; Aspirin Childrens [OTC]; Aspirin EC Adult Low Dose [OTC]; Aspirin EC Low Strength [OTC]; Aspirin Low Dose [OTC]; Aspirtab Maximum Strength [OTC] [DSC]; Bayer Aspirin EC Low Dose [OTC]; Bayer Aspirin [OTC]; Bayer Low Dose [OTC]; Bufferin Extra Strength [OTC]; Bufferin [OTC]; Childrens Aspirin Low Strength [OTC] [DSC]; Durlaza [DSC]; Ecotrin Maximum Strength [OTC] [DSC]; EcPirin [OTC] [DSC]; FT Aspirin Low Dose [OTC]; FT Aspirin [OTC]; FT Enteric Coated Aspirin [OTC]; GoodSense Aspirin Adult Low St [OTC] [DSC]; GoodSense Aspirin Low Dose [OTC]; GoodSense Aspirin [OTC]; Medi-First Aspirin [OTC]; Medique Aspirin [OTC]; Miniprin Low Dose [OTC] [DSC]; St Joseph Aspirin [OTC]; St Joseph Low Dose [OTC]; Stanback Headache Powders [OTC] [DSC]; Tri-Buffered Aspirin [OTC]; Vazalore [OTC]

Last Updated 12/13/24